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Development of a Locally Manufactured Toothbrush for School Children in Rwanda

Designing a sustainable intervention for school-based oral health

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Abstract

In Rwanda, dental health status is closely associated with socio-economic status, such as income level and place of residence. Toothbrush ownership varies significantly between urban and rural residents, and the vast majority of available toothbrushes are imported plastic products. An interdisciplinary project between KTH Royal Institute of Technology, the University of Rwanda, and practicing dentists aims to improve oral health in Rwanda through a school-based oral health program including fluoride rinsing, oral health education, and distribution of dental care kits. This master's thesis aimed to support that initiative by designing a user-friendly, locally manufactured toothbrush for primary school children and an implementation roadmap for the proposed intervention. This project was also aligned with Rwanda's national goals supporting preventive oral care, local production, and reduced dependence on imported plastic products.

The work was structured around an adapted Double Diamond design framework, integrated with principles from Design for the Base of the Pyramid and User-Centered Design, combining literature studies, market analysis, interviews, site visits, and questionnaires with school children. Literature studies and concept development were carried out in Stockholm while interviews, site visits and questionnaires were conducted on-site in Rwanda.

The result is a proposal for two types of bamboo toothbrushes including manufacturing outlines and an implementation roadmap for the intervention, from raw material sourcing to distribution in schools. Both toothbrush concepts are made from locally sourced bamboo with a beeswax coating and Nylon 11 bristles. The first concept prioritises manufacturing simplicity by utilising already available tools and skills in local workshops, with minimal initial investment. The second is a two-piece design with a replaceable brush head and integrated case, intended for larger-scale production. The thesis contributes a context-specific product system proposal that can support further development and planning of a school-based oral health program and local toothbrush production in Rwanda.

Keywords
Oral health intervention, School-based oral health program, Product system, Rwanda, Base of the Pyramid, Local manufacturing, Toothbrush design, Sustainable product design.

Sammanfattning

I Rwanda är tandhälsan tätt sammankopplad med socioekonomiska faktorer såsom inkomstnivå och bostadsort. Hur många invånare som äger en tandborste varierar kraftigt mellan städer och landsbygd, och de flesta tillgängliga tandborstar är importerade samt gjorda av plast. Ett tvärvetenskapligt projekt mellan KTH Kungliga Tekniska Högskolan, University of Rwanda och praktiserande tandläkare syftar till att förbättra munhälsan i Rwanda genom implementation av ett skolbaserat munhälsoprogram som inkluderar fluorsköljning, munhälsoutbildning och utdelning av tandvårdskit. Detta examensarbete syftade till att stödja detta initiativ genom att designa en användarvänlig, lokalt tillverkad tandborste för barn i grundskolan, samt en implementeringsplan för den föreslagna interventionen. Projektet var också i linje med Rwandas nationella mål om att stödja preventiv tandvård, lokal produktion och minskat importberoende av plastprodukter.

Arbetet strukturerades kring ett anpassat Double Diamond-ramverk, integrerat med principer från Design for the Base of the Pyramid och användarcentrerad design, och kombinerade litteraturstudier, marknadsanalys, intervjuer, platsbesök och enkäter med skolbarn. Litteraturstudier och konceptutveckling genomfördes i Stockholm medan intervjuer, platsbesök och enkäter genomfördes på plats i Rwanda.

Resultatet är ett förslag på två typer av bambutandborstar med tillhörande tillverkningsupplägg, samt en implementeringsplan, från råvara till distribution i skolor. Båda tandborstkoncepten är tillverkade av lokalt odlad bambu, behandlat med bivax och nylon 11-strån. Det första konceptet prioriterar enkel tillverkning genom att använda redan tillgängliga verktyg och kompetens i lokala verkstäder, med minimal initial investering. Det andra konceptet är en tvådelad design med utbytbart borsthuvud och integrerat fodral, avsett för storskalig produktion. Examensarbetet bidrar med ett förslag på kontextspecifikt produktsystem som kan stödja vidareutveckling och planering av ett skolbaserat munhälsoprogram och lokal tandborstproduktion i Rwanda.

Nyckelord
Oral hälsa, Tandhälsa, Skolbaserat munhälsoprogram, Rwanda, Bambutandborstar, Lokal tillverkning, Preventiv tandvård, Design for the Base of the Pyramid, Hållbar design, Teknisk design

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AI disclosure

In this project, AI has been used as a tool to improve the readability and language of the report. All text has been written by the authors and the language support from AI has consisted of prompting ChatGPT with original text and specific instructions of improving aspects such as shortening sentences, improving sentence structure, and clarifying wording in some parts of the report. Additionally, Leonardo.Ai was used to generate some images, which is explicitly stated when the image is presented.

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Nomenclature

UCD	User-Centered Design
BoP	Design for the Base of the Pyramid
NOHSP	Rwanda's National Oral Health Strategic Plan
MiR	Made in Rwanda
CBHI	Community-Based Health Insurance

1. Introduction

This chapter presents the background of the project, as well as a problem definition, purpose and aim. It also describes the methodology used throughout the project.

1.1 Background

Oral health is not only crucial for central body functions such as eating, speaking and breathing, it has a significant effect on life quality as well as the immune system. Today, half of the world's population suffer from dental caries and other oral diseases [1]. In Rwanda, both the common knowledge about oral health and the access to dental care are limited, especially for people living in rural areas [2]. Oral health services have historically been focusing on treatment of already existing oral diseases (reactive care) [3]. In Rwanda's National Oral Health Strategic Plan 2019-2024, it is emphasized that more resources should be devoted to preventive care through daily toothbrushing, fluoride toothpaste and school-based oral health programs [3].

In Rwanda, dental health status is closely associated with socio-economic status, such as income level and place of residence. Access to toothbrushes and oral hygiene knowledge varies significantly between urban and rural populations and economic constraints are reported as the main reason why many people haven't visited an oral health provider [2][4]. Since oral care is provided in district hospitals and not in the local health centres, rural inhabitants often have to travel long distances to receive treatment, which may result in travel costs and loss of income [1].

An interdisciplinary project between KTH Royal Institute of Technology, the University of Rwanda (UR) and practicing dentists aims to improve the oral health nationally in Rwanda by mapping the level of fluoride from the drinking water as well as introducing school based oral health programs. The school based oral health program would include fluoride rinsing, distribution of dental care kits and oral health education. This proposal aligns with WHO's Global Strategy and Action Plan on Oral Health 2023-2030 by introducing fluoride rinsing programs in schools to prevent cavities, expanding access to basic oral health tools like toothbrushes and fluoride toothpaste and prioritizing prevention, especially for children [5]. This master's thesis was developed to complement the implementation activities for the UR-KTH project and will focus only on parts of the overarching interdisciplinary project, specifically the dental care kits. The primary focus is the development of a user-friendly and locally manufactured toothbrush that can be distributed as part of the school based oral health program.

The Rwandan government is working towards making the country more self-reliant and environmentally sustainable. This is primarily done through two parallel strategies, the "Made in Rwanda" (MIR) policy aiming to promote local industry, and strict bans on single-use plastics to promote a circular economy [6][7]. However, the transition to local production and more environmentally friendly alternatives is financially and resource-demanding, presenting challenges for local companies lacking financial capacity to invest in new technologies or locally manufactured alternatives [8]. In addition, the national recycling industry remains underdeveloped with limited infrastructure and capacity for efficient separation of different materials [8].

1.2 Problem definition

There are significant differences in access to oral health care across Rwanda's population, and rural communities are particularly affected by limited availability of toothbrushes and a lack of oral health education. The lack of toothbrushes and education leads to high caries rates and other health risks.

1.3 Purpose

The implementation of a school based oral health program supports Rwanda's shift toward preventive oral health care, accessible across all socio-economic levels. The school based programs will provide children with appropriate oral hygiene tools, such as toothbrushes. However, the current dependence on imported plastic toothbrushes and limited recycling capacity create economic and environmental challenges [9]. This highlights a gap in locally manufactured, sustainable toothbrush alternatives suited to the Rwandan context. By developing a toothbrush adjusted to local production capabilities and government strategies, the purpose of this project is to develop an oral health intervention to support the implementation of preventive oral health strategies while contributing to long-term economic and environmental sustainability.

1.4 Aim

This project aims to develop a proposal for an oral health intervention in the form of a sustainable, locally manufactured, cost-effective and user-friendly toothbrush for distribution to school children as part of a national school based dental health program.

In order to realize this aim, the following four objectives have been formulated:

- To identify and evaluate locally available, sustainable, and economically feasible materials suitable for the production of a toothbrush within a local manufacturing context in Rwanda.

- To develop a proposal of a locally adapted manufacturing process for producing toothbrushes using the selected materials, taking technical feasibility, scalability, and cost-efficiency into account.

- To design a user-friendly toothbrush for schoolchildren in Rwanda, based on the selected materials, insights from user studies, and proposed manufacturing process.

- To outline the product system surrounding the toothbrush, covering material sourcing, manufacturing and implementation within the school-based oral health program.

1.5 Methodology

This section describes the methods and conceptual framework that guided the project.

1.5.1 Conceptual framework

This project was methodologically grounded in an adapted Double Diamond design framework, integrated with principles from Design for the Base of the Pyramid (BoP) and User-Centered Design (UCD).

The Double Diamond process was developed by the British Design Council 2004, and is a way of illustrating the diverging and converging phases of the design process as two connected diamonds [10]. In this project, the work is divided into five packages with separate focus areas, methods and deliverables. The first diamond includes three packages, each of which has an individual discover and define phase; "Literature study", "Field work" and "Analysis". The second diamond contains two parts; "Concept development" and "Result & Implementation", each having an individual develop and deliver phase. Figure 1 illustrates the workflow and the methods included in each diamond. Section 1.5.2 describes the methods included.

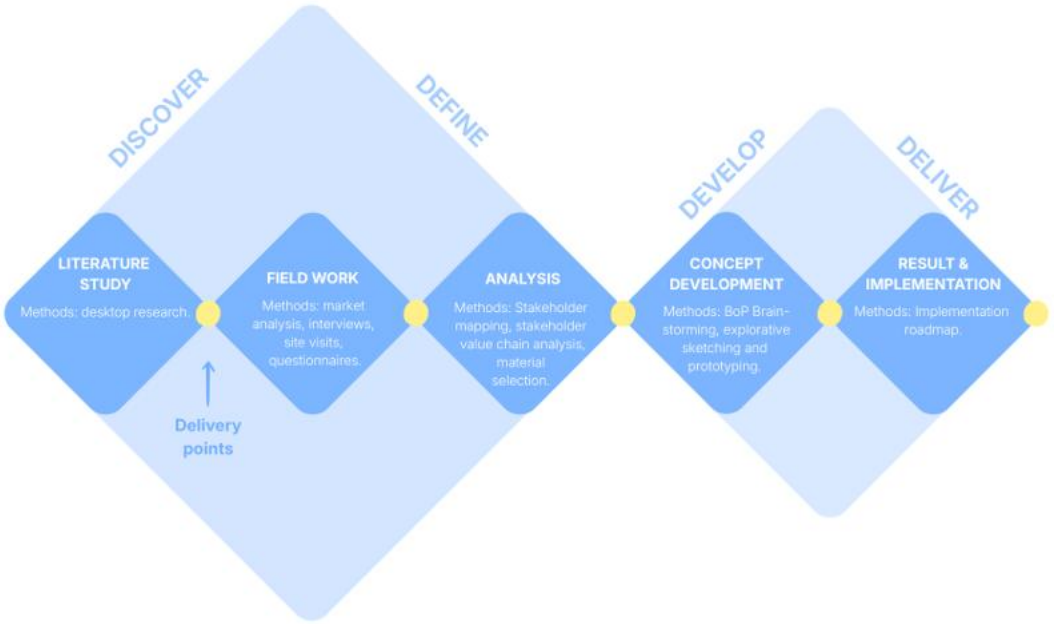


Figure 1. Visualization of the adapted double diamond workflow and the methods used in each diamond.

The Design for the Base of the Pyramid framework emphasizes the needs and barriers of low-income populations. Designing within BoP promotes affordable material choices, understanding of cultural contexts, simplified manufacturing processes and local value creation, which aligns well with the project aim [11]. This ensures that the proposed toothbrush is not only environmentally, but also economically and socially sustainable within the Rwandan context.

User-Centered Design (UCD) complements the BoP framework by focusing on the user's behaviours, preferences, ergonomics, and daily routines [12]. However, since UCD in itself does not address the systemic economic constraints and scalability, it is best used in combination with the BoP framework to make sure the result is desirable, but also feasible in a low resource context.

1.5.2 Methods

In this section, the methods used in the project are described.

Desktop research

Desktop research was conducted to build a foundation of knowledge across the key areas of the project. Academic articles, reports and official documents covering topics such as oral health, the Rwandan healthcare system, school based programs, toothbrush design and different materials, were reviewed.

Interviews

Semi-structured interviews were conducted with a range of stakeholders and knowledgeable persons in relevant fields. Semi-structured interviews are interviews where questions are prepared in advance in an interview guide, while allowing for the flexibility of asking spontaneous follow-up questions to gain a deeper understanding of topics that emerge during the interview [13]. Most interviews were conducted in person, however a few interviews were conducted online due to long distances or practical constraints. Before each interview, an interview guide with open ended questions was prepared to cover the most important areas. As some information on Rwanda-specific topics, like local value chains and local access, was outdated or not documented and available online to investigate during the desktop research, the interviews also filled in gaps from the first research phase.

Site visits

Site visits involve travelling to and observing a location, providing insight that cannot be obtained through interviews or literature alone. Two site visits were conducted during the field studies in Rwanda. The first was to the Relief and Development Support organization in Kayonza, including visits to two local schools, providing insight into the context the oral health program would be implemented in. The second was to a bamboo plantation and workshop in Musanze, organized by ARECO Rwanda Nziza, where material availability and existing production capabilities were investigated.

Questionnaires

Structured questionnaires were used to collect quantitative data about user preferences and habits. Structured questionnaires consist of questions with multiple pre-defined answers, which makes the questionnaire easier to fill out for the participant than a questionnaire with open-ended questions. Structured questionnaires also allow for easier quantification of the result [14]. Questionnaires were distributed to 242 schoolchildren at one public and one private primary school in Rwanda to gather quantitative data on current oral health habits, toothbrush usage, and preferences. The questionnaire consisted of 16 multiple choice questions formulated to be easy to understand for children, and was translated from English to Kinyarwanda with the help of a local translator. The structured format allowed for data to be collected from a larger number of respondents, providing a representative picture of the children's behaviours and attitude.

Stakeholder mapping
Stakeholder mapping is a method that helps identifying and visualising the key actors in a system. The stakeholders are identified and placed in a plot with "interest" on the x-axis and "influence" on the y-axis. Stakeholder mapping is an effective way of analysing the level of different stakeholder's importance, and how to manage them in a complex project [15]. In this project, a stakeholder map was created to gain a deeper understanding of the different stakeholder's roles, and how closely they should be managed.

Stakeholder value network
A stakeholder value network analysis was conducted by mapping the identified stakeholders in a network illustrating the flow of received and contributed value within the proposed system. The analysis was inspired by the value network analysis methodology described by Allee [16]. In value network analysis, value is described as both tangible exchanges such as money and materials, and intangible exchanges such as knowledge and services. The method was adapted to the context of this project, where the identified stakeholders of the proposed oral health intervention and their respective value exchanges were mapped. This helped clarify stakeholder roles, incentives, and interconnectedness within the proposal.

BoP Brainstorming
Brainstorming is a common design method used to generate a large number of ideas in a short amount of time. The participants are usually encouraged to call out as many ideas as possible without judgement or evaluation, prioritising quantity over quality [17]. In this project, an adapted brainstorming method guided by themes and strategies from the Design for Base of the Pyramid framework was used to generate ideas early in the concept development phase. The method was used in four sessions, where each session focused on generating ideas on a different theme and set of strategies. This method helped ensure that ideas were valuable and feasible for low-income contexts, rather than ideas better suited to higher-resource contexts.

Explorative sketching and prototyping
Explorative sketching and prototyping are methods commonly used to visualize, evaluate and test concepts [18]. In this project, explorative sketching was used iteratively to further visualize and evaluate ideas generated in the BoP Brainstorming session. To get a better idea of shape, feel and function of selected concepts, simple prototypes were made.

Implementation roadmap
Roadmapping is a visual planning method used to connect strategic objectives with the actions, resources, and steps needed to achieve them [19]. In this project, the method was adapted into a simplified implementation roadmap for the proposed product system. Rather than mapping technological development over time, as commonly done in technology roadmapping, the roadmap was used to outline the main actors and activities involved in moving the toothbrush from material sourcing and manufacturing to distribution and use in schools. The roadmap helped visualise the overall implementation logic of the proposal and identify key steps required for the system to function. The stakeholders and activities were based on the stakeholders and value exchanges identified during the analysis of the stakeholder value network.

1.6 Delimitations

This project aimed to deliver a proposal of a toothbrush designed for primary school children in Rwanda, including a manufacturing proposal. However, no physical production was set up within the scope of the project.

Prototypes were produced, but on a small scale with the machines available in the prototype workshop at Above Agency and KTH in Stockholm. No long-term durability, or clinical testing, was done within this project.

A proposed roadmap for the intervention was created, but a detailed curriculum for the school program was not developed since we, as design engineering students, lacked the competence in pedagogy and dentistry needed to make a feasible proposal.

Economic analysis like cash flow projections and break-even analyses were not done since many of the figures involved were not reliable enough at this stage. However, estimations of initial investments for the machines required for setting up the production were made.

Because of the language barrier and the young age of the target users, primary school children, no in-depth interviews were conducted with the target group. Instead, quantitative data was collected through questionnaires with questions formulated to be easy to understand for young children, translated to Kinyarwanda with the help of a local translator.

1.7 Ethical considerations

As Swedish students working in a Rwandan context it is important to be aware of the power imbalance that exists when external parties design solutions for communities that are not their own. Our cultural backgrounds, significantly different from the members of the target communities, inevitably colours our assumptions and perspectives. Before leaving Sweden, we participated in a preparatory workshop arranged at KTH that discussed these topics. Throughout the field studies, we aimed to include as many local perspectives as possible through involving a wide range of stakeholders in the process. However, we are aware that the risk of making biased decisions and assumptions cannot fully be eliminated in this kind of project.

For the on-site data collection, an ethical approval from The University of Rwanda College of Medicine and Health Sciences Institutional Review Board (IRB) was obtained. Since our data collection involved children, a vulnerable group, informed consent was sought from teachers, and assent was obtained from the children themselves before participation. Participants were informed about the purpose of the study, procedures involved, and their right to withdraw at any time without any consequences. All data collected during the study was kept confidential, and the questionnaire responses were anonymized.

2. Literature study



In the literature study-diamond, desktop research was conducted to build a foundation of knowledge of the key areas of the project. The current oral health situation, including the oral health care system, was examined to understand the context the program will be implemented in. Government policies and national strategies were reviewed, as the Rwandan government is both a key stakeholder and primary customer of the toothbrushes. To draw inspiration from previous school programs, previous programs in both Sweden and Rwanda were investigated. Rwanda's existing recycling and waste management infrastructure was investigated, to collect information for end-of-life considerations for the toothbrush. A review of ISO standards relevant for toothbrush manufacturing was also conducted. Finally, a thorough investigation of different potential materials and manufacturing methods for the toothbrush was conducted to support material selection and manufacturing proposal development later in the project.

2.1 Oral health in Rwanda

Tooth and periodontal diseases are among the top 10 causes of morbidity in all age groups in Rwanda, and 4th deadliest among people over 5 years. Risk factors caused by lifestyle changes such as an increased consumption of sugars, alcohol and tobacco have increased in recent years due to urbanization and heavy marketing. [3]

In 2016, the first (and most recent), national oral health survey was conducted in Rwanda. The results of the survey demonstrated a high prevalence of untreated dental caries across the nation and limited access to oral hygiene tools and preventive practices. Almost two-thirds of the population had experienced tooth pain in the past year, and 60% of the population aged 20 years and up had caries, while 70.6% had never visited an oral health provider. [2]

Dental health status is closely associated with socio-economic status such as income level and place of residence. While 71% of the urban population own a toothbrush, the corresponding percentage is 25% in rural areas. Another issue is the limited awareness of the importance of good oral health, and lacking knowledge of healthy oral hygiene habits. A toothbrush design that is intuitive and easy to use is of priority to decrease hurdles for users with little prior experience and knowledge. [20]

2.1.1 Oral health care in Rwanda

Oral healthcare in Rwanda focuses primarily on treatment of acute symptoms (reactive care) rather than prevention. In the National Oral Health Survey Rwanda 2016, 98.7% of respondents who had sought oral care did so for pain relief. In 2019, tooth and gum disease was the most common complaint among patients visiting district hospitals [2]. This suggests a high burden of oral disease and indicates that care is often sought once symptoms become severe. In 2018, there were only 228 oral health workers in the country, which equals 0.2 oral health workers per 10 000 inhabitants [21].

The Rwandan health care is divided into four levels:

- **Primary level: Health Centers (HC).**
  Local health centres are the primary health care level, staffed by community health care workers living in close contact with the population. In the Health Centers, oral health services are only provided in the form of analgesics (pain relief) and there is generally no skilled oral health staff [3].

- **Secondary level: District hospitals**.
  District hospitals receive cases referred from health centers, for example for basic oral care like caries treatment and infection management. In district hospitals there are normally at least two dental therapists, but equipment such as dental chairs are often lacking or not properly working due to limited maintenance capacity [3].

- **Tertiary level: Provincial hospitals**.
  Provincial hospitals receive more complex referrals from the primary or secondary level. Provincial hospitals provide specialised treatment and are often staffed by dental surgeons [3].

- **National referral hospitals**.
  National referral hospitals are highly specialised hospitals or university hospitals. These hospitals are equipped to handle advanced procedures such as maxillofacial surgery [3].

With 87-91% of the population covered by health insurance, Rwanda is the country in Sub-Saharan Africa with the highest insurance coverage rate. The majority of the population, 81.6%, is enrolled in the Community-Based Health Insurance (CBHI). The insurance premium is based on the household's financial status through a system called Ubudehe. The fee is completely subsidized for the poorest residents and 3 000 RWF (approximately 2 USD) per person and year for the majority, while higher income residents pay 7 000 RWF (approximately 5 USD) per person and year [22]. CBHI covers basic oral healthcare, such as medical consultations, treatment for common oral diseases and most priority medicines for oral health. Despite this, 70.6% of the population had in 2016 never visited an oral health provider, and many patients decide to discontinue treatments when referred to higher-level hospitals. For rural residents, a referral to a district or provincial hospital for treatment often requires transportation and accommodation costs, and results in loss of work income that may be crucial for a lower-income family [3].

2.2 Government strategies and policies

As the Rwandan government is an important stakeholder in the proposed oral health intervention, relevant national strategies and policies were reviewed. Current oral health strategies aim to shift oral care from reactive treatment to proactive prevention. Other national policies focus on increasing self-reliance and improving environmental sustainability, mainly through the Made in Rwanda policy and the plastic ban.

2.2.1 National Oral Health Strategic Plan

Rwanda's National Oral Health Strategic Plan (NOHSP) 2019-2024 aims to decrease the burden of oral and dental diseases and improve the overall oral health of the population. The plan emphasizes the importance of a shift in focus from reactive treatment toward preventive care through daily toothbrushing, fluoride toothpaste and school-based oral health programs. Introducing oral health programs has proven effective in previous projects, for example by an initiative led by SOS Children's villages, supported by Danish Dentists Without Borders, that demonstrated a decrease of visible dental caries from 45% to 16% among school children after the implementation of a school based oral health program. The strategic plan also emphasizes the importance of including oral care in the primary health care levels to increase the availability, particularly in rural areas. The plan involves equipping local health centers with the necessary tools and providing training and education for health staff to deliver basic oral health services. By increasing the availability of preventive dental care and promoting healthy oral habits from an

early age, Rwanda hopes to decrease the need for expensive emergency care and improve the long-term oral health and general wellbeing across the population [3].

2.2.2 Made in Rwanda

Though Rwanda is a fast-growing developing country, the industry sector is still small and many companies struggle to compete globally. Most companies are small or medium-sized and the access to funding, raw material and competence is limited. The Made In Rwanda policy was launched in 2017 to tackle Rwanda's current dependency on import and increase the competitiveness of the Rwandan economy. This is done by strengthening local manufacturing through several measures, for example by introducing tax exemptions on machinery and raw materials and granting local producers a 15% preference in public procurement, as long as they can demonstrate at least 30% local value addition [6].

2.2.3 Plastic ban

Rwanda has gained international recognition for their strict plastic policies, including a ban on most single-use plastics and implementation of R-principles (for example Reduce, Reuse, Recycle). However, the transition to more environmentally friendly alternatives is financially and resource-demanding, presenting challenges for local companies lacking financial capacity to invest in new technologies or locally manufactured alternatives to plastic packaging [3].

There have been two instances where the manufacturing, use, importation or sale of single-use plastic was prohibited in Rwanda, the first of which was in 2008 and specifically targeted single-use plastic bags. The other ban was in 2019, which was a broader ban on single-use plastic, on items such as straws, plastic bottles, plastic utensils and plastic packaging. There are however exceptions to the prohibition. To be exempt of the prohibition, an application has to be submitted, and accepted by the competent authority. In the case where the application is successful, the competent authority establishes guidelines on procedures and conditions for the issue of exceptional authorisation. Imported goods packaged in plastic material or single-use plastic items are subject to an environmental levy in accordance with relevant laws. These regulations cover single-use plastics and plastic packaging, and do not cover plastic items meant for multiple uses, for example toothbrushes [7].

2.2.4 Bamboo sector support

The promotion of bamboo is integrated in national goals such as the Made in Rwanda initiative, the National Strategy for Transformation and the Green Growth and Climate Resilience Strategy [23]. These policies aim to expand the bamboo sector from low-value handicrafts toward more industrial manufacturing to achieve import substitution for wood and fiber products. Furthermore, the government promotes cultivation to ensure a consistent and sustainable supply for raw materials, thereby supporting the development of bamboo-based industries. This is largely due to the benefit of planting bamboo in buffer zones along rivers and on degraded land, where it contributes to erosion control and ecosystem restoration [23]. These measures indicate a policy environment that is conducive to investment in manufacturing of bamboo.

2.3 Recycling in Rwanda

Recycling capabilities in Rwanda are growing but are currently underdeveloped and are lagging when compared to the increase of waste generated. In Kigali, waste increased from 180 tonnes per day in 2012 to 500-800 tonnes per day in 2016. Despite waste management constantly being developed, the pace of which waste was being increasingly generated was not translated to waste management capabilities of the same scale [24]. In 2008 there were no recycling companies in Rwanda, whereas in 2019 there were 14, exemplifying the growth of recycling in Rwanda [8].

The recycling in Rwanda is currently highly selective in terms of what is being recycled, and the recycling is hampered by the fact that there is no effective method to divide materials. It is stated that "Currently all recycling is from mixed wastes, either by collection workers employed by the private companies, or by pickers working at Nduba or at illegal dumpsites.", which reduces the quality and the feasibility for a larger scale of recycling of plastics in Rwanda [25].

It is difficult to pinpoint the exact recycling rate in Rwanda as different sources cite different percentages. One source states the overall recycling rate in specifically Kigali being around 2 % [26], whereas another source also mentions the 2%, but states that a more accurate number is 12% based on their MFD (material flow diagram) [25]. There are also different numbers in regards to the recycling rate specifically for plastic in Rwanda. According to the news organization "Downtoearth", the recycling rate for plastic in Rwanda is 70-80% [27], and is cited as a source by the World bank [28]. Another source estimates the recycling rate of plastic in the entirety of Rwanda being around 2-3% [29].

Something that several of these sources state is that the current capabilities of recycling in Rwanda are limited, and that the information regarding the recycling rate is limited. Given this, the project should strive for a toothbrush design with biodegradable materials and materials with low environmental impact when the material is produced/cultivated.

2.4 School-based health programs

School-based health programs were studied to understand the impact and feasibility of such programs.

2.4.1 School based oral health program in Sweden

An example of a preventive, nationwide oral health action is Sweden's nationwide school oral programme in the 1960s, which resulted in a dramatic decline in childhood caries by integrating daily brushing, fluoride rinsing and oral health in the education system [30]. Applying similar principles in Rwanda by providing dental care kits and integrating oral hygiene routines in school education, aligns with the NOHSP. A school-based oral health program has strong theoretical potential to reduce disease burden and increase access to dental care across all income levels.

2.4.2 School based feeding program in Rwanda

Another example of a school based programme is the school feeding programme that was recently implemented in Rwanda - "National School Feeding Programme" (NSFP) [31]. It was introduced on a large scale in 2019 and has expanded since then, with a nationwide coverage for pre-primary school children up to secondary school children since 2021. The age range included in pre-primary to secondary school are 3-18. NSFP includes providing kitchenware to schools and subsidies from the Rwandan Government for the cost of food. The subsidy is set at 219 RWF (approximately 0.15 USD) for primary school children and 642 RWF (approximately 0,44 USD) for secondary school children. If the cost of the meal is higher than this, the parents will pay the difference between what is contributed by the NSFP and the actual price of the meal. Since the introduction of NSFP, students' health and attendance has improved while dropout has reduced. This feeding program aligns with the Education Sector Strategic Plan (ESSP 2024 - 2029) as it, among other goals, aims to provide school children with a nutritious diet [32]. The (ESSP 2024 - 2029) also aims to improve children's health and improve knowledge regarding health. The UR-KTH project, which aims to improve children's oral health thus also aligns with the (ESSP 2024 - 2029). The feeding program was initially started in 2002 and was then led by the World Food Programme WFP, which provided the Ministry of Education in Rwanda with technical and financial support. As mentioned it has grown significantly, and in 2014 WFP's "Home-Grown School Feeding" program was implemented. It was primarily funded by WFP but during implementation they worked in close collaboration with the Ministry of Education. The Home-Grown School Feeding program initially targeted 140 schools across seven specific districts with high stunting rates [31]. This program shows that a nationwide school-based program is feasible to implement and that the Ministry of Education would likely be interested in helping implement such a program.

2.5 Toothbrush design

Key aspects of toothbrush design were reviewed to inform the development of an effective toothbrush, based on established design principles and standards.

2.5.1 Effective toothbrush design

The significance of the design and configuration of toothbrush bristles in relation to the effectiveness of plaque removal has been widely studied, but the results from these studies are mixed. Multiple studies show that toothbrushes with multi-level, angled or cross-bristle configurations can improve plaque removal when compared to a flat surfaced bristle configuration [33][34]. According to these sources, these configurations are better at reaching areas where plaque accumulation often is more difficult to remove, specifically interproximal spaces and along the gingival margin.

In contrast, other sources have not been able to show that bristle configuration has any statistically significant differences on plaque removal. Flat surfaced bristle configurations are comparable to more complex designs in terms of plaque removal [35]. Other sources also mention that brushing technique and other user-related factors has a more significant effect on plaque removal rather than the bristle configuration [36, 37]. It is also explicitly stated that "The

three brushes were capable of effectively removing dental biofilm. The arrangement and size of the bristles had no influence over oral hygiene. Thus, there is no justification for using more expensive toothbrush models on schoolchildren. This point is especially important in communities where cost is a relevant factor." [37, p. 153]. This refers to one toothbrush with bristles on the same plane with straight arrangement, one with bristles on different planes with straight arrangement and one with bristles on different planes and both straight and circular arrangement. This suggests that there is little difference in plaque removal when comparing the arrangement of bristles of different toothbrushes and that a more simple arrangement is suitable for the context of this project.

Another design factor for toothbrush bristles is the stiffness. In a systematic review of bristle stiffness and end-shape related to adverse effects on soft tissues during toothbrushing, it was explained that while harder bristles have a better ability to remove plaque, they also cause more soft tissue trauma (gingival lesions and increased bleeding) [38]. The systematic review suggests that soft and extra soft toothbrushes tend to be more safe as they cause less oral damage. However, what stiffness is optimal is dependent on the user's current oral health [39]. Soft bristled toothbrushes are recommended for users already showing signs of soft tissue damage. Harder bristles may provide improved plaque removal for some users with poor oral hygiene, although they are associated with increased soft tissue trauma. Medium stiffness bristles may be suitable for some users, but they still present a greater risk of soft tissue irritation compared to soft bristled toothbrushes.

2.5.2 ISO standards

The standards ISO 20126, ISO 7405 and ISO 10993 have been reviewed to design the toothbrush according to safety compliance. ISO 20126 (Dentistry - Manual toothbrushes - General requirements and test methods) addresses mechanical performance and physical safety of manual toothbrushes, ISO 7405 (Dentistry - Evaluation of biocompatibility of medical devices used in dentistry) and ISO 10993-1 (Biological evaluation of medical devices Part 1: Evaluation and testing within a risk management process) address biological evaluation.

Toothbrushes are classified as devices with limited contact with oral mucosa. The primary biological safety requirements therefore include cytotoxicity, irritation and sensitization [40]. The finished product is assessed in terms of safety, where materials, manufacturing processes and potential contaminants or additives are taken into account. Biological evaluation must be integrated into a structured risk management process, ensuring that material selection and production methods do not introduce harmful substances during use [41].

ISO 20126 provides general requirements for durability and safety of manual toothbrushes [42]. The requirements include resistance to breakage, structural integrity and tuft retention. These requirements inform the need for a robust and mechanically stable design.

2.6 Bristle material

Historically, many different fibres have been used to clean teeth, for example tree fibres and boar hair [43]. Ever since nylon bristles were launched in America in the 1930's, synthetic fibres have gradually replaced natural ones thanks to their ability to be manufactured with uniform diameters and have rounded, gentler bristle edges. Today, the market is dominated by nylon-bristled toothbrushes [44]. To find a bristle that both could comply with standards and requirements regarding teeth cleaning, and with the plastic regulations in Rwanda, different bristle materials were investigated.

2.6.1 Natural fibres

Chewing sticks

Chewing sticks made from roots, twigs or stems from a variety of plants have been used for cleaning teeth for thousands of years and remain frequently used in many low- and middle income countries. Salvadora persica, also called miswak tree, is the most commonly used species [45]. Salvadora persica is not currently cultivated in Rwanda but is native to neighboring Uganda and Tanzania, which suggests that cultivation in Rwanda may be possible [46]. The use of chewing sticks is endorsed within islamic teachings and serves not only as an oral health tool but also as an expression of cultural identity and religious devotion, and is thus particularly frequent in muslim-majority regions [47].

Studies comparing the efficiency of chewing sticks and conventional toothbrushes and toothpaste show that the plaque removal and gingivitis prevention are comparable, and in some cases chewing sticks performed significantly better than toothbrush and toothpaste [45]. However, the maneuverability of a chewing stick is reduced compared to that of a conventional toothbrush. Because the fibers are aligned along the longitudinal axis, it is hard to reach the inside of the teeth. Safe and effective use of the chewing stick also requires appropriate technique since brushing too hard, infrequent trimming of the fibres or using a dried out chewing stick could potentially be harmful [48].

The use of chewing sticks is not widespread in Rwanda. One study reported that 1.6% of the adult population used chewing sticks, compared with 12% who reported using charcoal or other alternative cleaning methods [20]. Though chewing sticks is a cheap, potentially locally producible and natural way of cleaning teeth, conventional toothbrushes are more commonly used and accepted in Rwanda. Promoting chewing sticks in this context may involve cultural hurdles and would require user education to ensure correct and safe technique.

Boar hair

Before the introduction of nylon toothbrush bristles in the 1930's, animal hair bristles dominated the market [43]. Thanks to their biodegradability, boar hair bristles have recently gained popularity among communities seeking more sustainable toothbrush alternatives or aiming to reduce exposure to microplastics. However, the porous structure of animal hair leads to a higher

water absorption compared to nylon alternatives, which may cause bacterial growth and reduce hygiene [49]. Animal hair may also raise ethical considerations associated with animal derived products, and be subject to cultural barriers to acceptance.

Other natural fibres
According to the market analysis in 3.1, toothbrushes with natural fibre bristles other than boar hair appear to be very uncommon. A study comparing the efficiency of plaque removal of ridge gourd, coconut, bamboo and conventional nylon fibres found that nylon bristles performed better than the natural alternatives. However, bamboo bristles were more durable and showed the least signs of wear after the equivalent of 90 days of normal toothbrush use in a tooth brushing simulator. In the study, only the nylon toothbrush and the bamboo bristled toothbrush were commercially available, the ridge gourd and coconut bristled alternatives were manufactured specifically for the study. The study did not specify which kind of bamboo fibres the bamboo bristled toothbrush had [50].

2.6.2 Nylon

Nylon bristles have dominated the toothbrush market for many reasons. Compared to natural fibers, nylon offers more efficient plaque removal, better resistance to bacterial growth and can be manufactured with higher precision which contributes to predictable performance. Nylon filaments are manufactured with rounded tips, which decreases the risk of soft tissue irritation and complies with the ISO standards for bristle ends [44].

Castor oil-based nylon has become increasingly popular among toothbrush companies aiming to offer more environmentally friendly products. Castor oil nylon is chemically very similar to conventional petroleum-based nylon, and therefore has comparable mechanical properties, including stiffness and flexibility. Studies have shown that castor oil nylon bristles can be manufactured in compliance with the current ISO standards for toothbrush bristles, and that the plaque removal efficacy is equivalent to conventional nylon bristles. However, though castor oil nylon is produced by renewable resources and often is associated with a lower carbon footprint than conventional nylon, it is still non-biodegradable [44]. Nylon is generally less recyclable than commodity plastics such as polyethylene (PE) and polypropylene (PP), as recycling of Nylon generally is a more complex and expensive procedure [51]. As a result, Castor oil-based nylon offers the possibility to manufacture bristles that complies with ISO standards with a lower dependence on fossil resources than conventional bristles, but it does not resolve end-of-life challenges and could contribute to microwaste.

2.7 Handle material

Since the handle makes up most of the volume of a toothbrush, it is both the main element conveying the visual appearance of the product, and has a large impact on the overall ecological footprint. Three main materials were investigated for the handle, bamboo, compostable bioplastic and different woods.

2.7.1 Bamboo

Toothbrushes where the handle is made of bamboo are steadily growing in popularity [52]. There are multiple reasons for why this material is chosen, the primary reason being its environmental sustainability. According to [53], the bamboo toothbrush shows a decrease of 68% in carbon footprint when compared with the traditional plastic toothbrush (in this example they assume that the production takes place in Switzerland, and that the bamboo is grown in and transported from China). Bamboo is also very fast growing, with some species of bamboo being able to grow as rapidly as 20 meters in 45-60 days [54]. It is also biodegradable, meaning that the handle of a bamboo toothbrush can degrade fully without leaving persistent waste at end of life [55]. When compared to plastic, bamboo toothbrushes also show that the hygienic condition was comparable to plastic toothbrushes [56].

There are many types of bamboo, and one of the most commonly used in manufacturing is moso bamboo (Phyllostachys edulis/Phyllostachys pubescens). It is also the most commercially important bamboo species in China in terms of cultivation and value added as more than 72% of bamboo forests in China are moso bamboo forest [57], with China accounting for approximately 75% of the global exports of bamboo commodities [58]. Moso bamboo is widely used as the primary material for bamboo toothbrush handles, including toothbrushes from companies such as PÄRLA [59], The Organic Republic [60], and Bamboovement [61]. This widespread use is due to the combination of its favorable physical, mechanical, and processing properties. Although some manufacturers do not specify the species of bamboo used, no toothbrush explicitly made from a species other than moso bamboo was identified during this project.

The advantageous physical properties of moso bamboo (Phyllostachys edulis) include its large, cylindrical culms, which provide a significant volume of usable material for industrial applications [62]. However, its geometry is not uniform, as both culm diameter and wall thickness decrease with increasing height on the plant, with one source citing that the diameter at the base part of a mature culm is around 16.8 cm and 12.4 cm closer to the top [63]. This source also mentions that the average wall thickness of a mature moso bamboo culm is 1.2 cm when measuring the thickness 130 cm above the ground. It is crucial that the thickness is large enough for a toothbrush to be carved out from it. moso bamboo performs well in terms of mechanical properties, possessing a tensile strength approximately twice that of wood [64]. While the material is very strong, it is strongest on the outer wall and weaker towards the inner wall [64], something that all bamboos have in common due to higher fibres density close to the outer wall [65]. This is important to have in mind for bamboo products in order to avoid using the weaker inner material.

The research conducted during this project was not able to identify any moso bamboo plantations near or in Rwanda. Additionally, the local climate is not well suited for its cultivation. Rwanda has a tropical highland climate with relatively stable temperatures around 16–20°C and above year-round [66], whereas moso bamboo typically thrives in a subtropical monsoon climate with hot, rainy summers and cold, dry winters, where temperatures can drop below 0°C [67]. For more certainty in regards to the cultivation of moso bamboo in Rwanda, moso bamboo could be planted in different parts of Rwanda to test its viability.

Species that grow in Rwanda were therefore identified during the field studies, and the factors that make moso bamboo favorable for toothbrush manufacturing was considered when evaluating alternative species. One of the most important properties are the dimensions, primarily thickness but also diameter, as a larger diameter offers more material, hence more toothbrushes per culm.

2.7.2 Wood

Eucalyptus wood makes up approximately 80% of the forestry sector in Rwanda [68]. Eucalyptus wood is characterized by high density and mechanical strength. It is commonly used as firewood, but also within furniture production and other wooden products. Eucalyptus wood is a fast-growing species which is an advantage in low-income areas where long-term investments in forestry may be financially challenging. Since it is regenerated naturally, less forestry expertise is required for maintaining an eucalyptus forest [69]. Eucalyptus is grown in large parts of the country which makes it easier to access without the need of long transportation distances [68].

2.7.3 Compostable bioplastics

Compostable bioplastics are plastic materials that partly, or completely are made of organic raw materials and naturally decompose within a time span comparable to other compostable materials without leaving any visually distinguishable or toxic residues [70]. Compostable bioplastics are also one of few exceptions from the Rwandan single-use plastic ban [71]. There is potentially good access to raw materials for production of bioplastics as 64-74% of the waste generated in Rwandan cities is organic waste, which means that there is a big un-utilized source of biomass in the country [9]. Arth Biobag Ltd is an example of a company currently manufacturing plastic bags out of compostable bioplastics in Rwanda. This means that compostable bioplastics potentially is a locally producible and sustainable material that could be used for manufacturing of toothbrush handles.

2.8 Toothbrush manufacturing

When starting production in a context where there currently is little industrial production, it is especially important that the initial costs and complexity of setting up production are low. Because of this, manufacturing methods for different materials were studied as this could impact the choice of material.

2.8.1 Bamboo and wood handle manufacturing process

There are multiple ways of manufacturing bamboo toothbrushes, depending on the shape of the toothbrush. The most common type of bamboo toothbrush handle are those with two flat surfaces on the front and back side, and more intricate shapes on its sides. This type of toothbrush is manufactured by splitting bamboo culms into smaller wedges, which are then treated, either with chemicals or heat. They are then planed to create rectangular bamboo planks and the planks are milled into the desired shape of the toothbrush. Holes are drilled into the head of the shaped plank, and bristles are inserted, and a logo is laser printed [72]. This process

is often made in the context of a large-scale production, where each step requires specialized, large machines.

Different methods adapted to a smaller scale production were analysed to enable a production with smaller initial investments. The overall process would be similar but most of the more expensive, automated machines used would be substituted to smaller machines requiring more manual labour. In the more conventional large scale production, the following machines are used: Sawing machine, bamboo splitting machine, bamboo strip planing machine, high precision bamboo strip planing machine, precise cutting machine, shape forming milling machine, polishing machine, drilling machine, toothbrush tufting machine, logo branding machine. For the smaller scale production with lower initial costs the machines used would be the following: Sawing machine, manual bamboo splitting machine, thicknessing machine, precise cutting machine, router table, sanding belt, drilling machine, toothbrush tufting machine, laser printing machine. More expensive machines that were required and could not be substituted were the drilling machine and toothbrush tufting machine. Different ways of treating bamboo were also investigated. Bamboo can be treated in different ways, one of which is boiling. Bamboo can be boiled in water under ordinary pressure in order to remove substances such as starch and sugar. This improves dimensional stability by reducing moisture absorption and improves decay resistance by removing sugar and starch. Bamboo can also be heated by pressurized steam. This method is better in terms of improving dimensional stability and decay resistance but requires more complex equipment. Whereas boiling only requires open-tank equipment as heater water baths, steam heat treatment requires sealed pressure vessels capable of maintaining elevated temperature and pressure conditions [73].

2.8.2 Compostable bioplastic handle manufacturing process

The manufacturing process for a compostable bioplastic handle involves relatively few steps, but requires a higher level of technical expertise and initial investments than bamboo and wood manufacturing. The first step of the process is to source bioplastic granules. These granules are melted in an injection molding machine, and injected under pressure into a steel or aluminum mold. After cooling, the mold is split and the solidified toothbrush handle is ejected [74]. Once the handle is produced, holes for bristles are drilled and the bristles are attached using a tufting machine.

The molds are typically made from steel or aluminum and designed with cooling channels that circulate water during the cooling process. These molds require CNC machining and are often one of the largest investments in the production setup. Injection molding machines also require a steady power supply, setting up a factory where the access to electricity is unreliable could cause inconsistent temperatures, pressure or in severe cases lead to molten plastic solidifying and damaging the machine [74]. The positive aspect of injection molding is that it is a relatively easy way to efficiently mass produce toothbrush handles, and after passing the break-even point of the initial investment, the unit cost becomes very low.

3. Field work

In this chapter, the methods and results within the “field work”-diamond are presented. The field work on-site in Rwanda included a market analysis, interviews and site visits with both private companies and public institutions and quantitative data collection in primary schools. The market analysis provided inspiration from already existing toothbrush models and insights in the current toothbrush market in Rwanda while the main purposes of the interviews and visits were evaluating material availability, investigating practical conditions for local production and gaining a deeper understanding for the context the school program will be implemented in. The quantitative data was collected through questionnaires and provided information on school children's current toothbrushing habits and preferences.

3.1 Market analysis

To gain inspiration from already existing toothbrushes, a selection of toothbrushes on the global market was analyzed. The Rwandan toothbrush market was also analyzed in order to better understand the context of the toothbrush market specifically in Rwanda.

3.1.1 Global market

The global toothbrush market is steadily expanding, driven by an increased awareness of, and interest in, oral health [75]. Although the global market is still dominated by conventional plastic toothbrushes, the market for bamboo alternatives is growing, especially in high income countries. The increase is mainly driven by a growing demand for plastic-free and environment friendly options [76]. Many toothbrush brands are offering toothbrushes designed to decrease plastic waste, either by offering toothbrushes with replaceable heads or handles made of recycled or natural materials. Some of the big toothbrush brands focusing on reducing plastic waste are The Humble Co, Brush with Bamboo and Bite, however, all plastic reduced options found during the market analysis were targeting high or middle income markets and had a higher price point than plastic alternatives. Figure 2 shows some of the toothbrush designs found during the market analysis.



Figure 2, toothbrushes designed to minimize plastic waste. From left to right: The Humble Co. (plant based plastic), Jordan (replaceable head), Bite (bamboo toothbrush with replaceable head), Brush with Bamboo (bamboo toothbrush and travel case), Preserve (recycled plastic from yoghurt cups), TePe (wood and plastic with replaceable head).

The most common packaging of the observed toothbrushes was a cardboard back and transparent plastic front. Transparent plastic is effective in showing the toothbrush in stores. Cardboard boxes are also common, in these cases the toothbrush is showcased on the cardboard through a photo. The Humble Co. sells their toothbrushes in cardboard boxes, the toothbrushes are however not loose in the cardboard box but are packaged in a kraft paper bag within the cardboard box.

3.1.2 Rwandan market

During the field work in Rwanda, the vast majority of the toothbrushes observed across the visited stores were imported and made of plastic. In total, 24 different stores were visited during the time spent in Rwanda, where 6 stores were located in rural areas and 18 located in urban areas, mainly in the capital Kigali. In Kigali, toothbrushes are normally sold in supermarkets and pharmacies. The price of manual adult toothbrushes found in local stores ranges from around 400 RWF (approximately 0.3 USD) for simple, basic toothbrushes, to 4 000 RWF (approximately 3 USD) for toothbrushes marketed as more advanced, with features such as charcoal infused bristles or "dual action" design. Only one toothbrush with a handle made of other materials than conventional plastic could be observed during the market analysis: a toothbrush from the Turkish brand Difas, seen in Figure 3. The product packaging claims that the toothbrush is made of "100% eco friendly and recyclable materials" and that the bristles are made of "100% bio based fibres". Information on the packaging and the manufacturer's website indicates that the handle is made of an olive-seed added plastic, and that the bristles are made of an unspecified bio-based material derived from renewable sources. This toothbrush may have a reduced amount of fossil-plastic due to the plant-based additives in the handle, but is not biodegradable and is unlikely to be recyclable in conventional waste management systems due to the mix of inseparable materials. A local plastic toothbrush manufacturer called Velano toothbrushes are currently operating in Kigali, but Velano toothbrushes could only be observed in one store, a supermarket located in the central parts of Kigali. No other locally manufactured toothbrushes were observed in any of the visited stores. Colgate was by far the most commonly observed brand, while the Turkish brand Difas and Corona Plus were also present in some stores.



Figure 3. Toothbrushes observed during the market analysis.

3.2 Interviews

All interviews were conducted using a semi-structured approach to allow for flexibility to explore all topics discussed thoroughly. All interview guides can be found in appendix A. The very first interview conducted was with ARECO Rwanda Nziza in Kigali. ARECO is a government funded non profit organisation working with ecological preservation, climate change adaptation and biodiversity conservation through engaging the local communities. The representatives from ARECO shared information on their ongoing bamboo planting projects in buffer zones, and a follow-up site visit was arranged to assess supply capacity and scalability more thoroughly (see section 3.3.2). ARECO showed interest in the project and offered to facilitate relevant contacts, or become a partner in the project if bamboo is selected as material for the toothbrush.

An interview with Ultimate Forests Company suggested that softer wood species, particularly pine, may be well-suited for smaller wooden components like toothbrushes, and provided contact to Rwanda Wood Value Chain Association (RWVCA). In an interview with RWVCA, these findings were strengthened, when the interviewee confirmed that eucalyptus, pine, and cypress already are used in smaller wooden product manufacturing. The interviewee also commented that bamboo is more cost-competitive than conventional timber and that large-scale bamboo manufacturing is possible, but there is potentially a need to further develop supply chains. They also offered to support the establishment of new plantations if necessary.

Through RWVCA, contact with the Masaka Incubation Center was established. The incubation center is supporting young entrepreneurs in Masaka, and hosts a bamboo workshop. During an interview held at the incubation centre, a representative from the organisation recommended the bamboo called dendrocalamus for toothbrush manufacturing. Dendrocalamus is not a native species, but is cultivated in the area and has properties comparable to the more commonly used moso bamboo. The representative from Masaka incubation center also provided contact to an entrepreneur within bamboo manufacturing that could act as a potential manufacturing partner.

Interviews with Arth Biobag, a Kigali-based company producing home-compostable plastic bags, confirmed that manufacturing toothbrush handles from home-compostable bioplastic is technically feasible. The company currently sources its granules from India due to high local production costs, but has plans to establish domestic bioplastic production in Rwanda within two years. They also expressed openness to collaboration, for example by sharing facilities and personnel, if bioplastic were selected as the material for the toothbrush manufacturing.

Interviews with institutional stakeholders indicated support for the bamboo industry and local production. The Rwanda Development Board (RDB) encouraged use of bamboo and emphasised supportive government policies for locally produced products, such as tax exemptions on machinery and raw materials. The Rwanda Environment Management Authority offered a more optimistic assessment of the national waste management system than the findings in the previous desktop research, stating that all collected waste is sorted, and that plastic waste is largely recycled. RDB also mentioned that most production that takes place in Kigali is located in Kigali special economic zone since infrastructure including roads, electricity and water is best

there in regards to production. It is also incentivized to produce in Kigali special economic zone by tax reductions and government support in relation to investment and industrial growth.

At SOS Children's Villages, a dentist responsible for the organisation's oral health projects, was interviewed. The interviewee described a recently launched school-based oral health initiative, though limited to a single school. The interviewee suggested that the lack of toothbrushing habits in rural areas mainly depends on lack of information, not limited access to tools.

3.3 Site visits

During the field studies in Kigali, two site visits were made, one to the Relief and Development Support Organization in Kayonza and one to a bamboo plantation in Musanze.

3.3.1 Relief and Development Support organization

During the visit to Kayonza, the Relief and Development Support organization was visited. The founder of the organization presented different ongoing initiatives, including an agricultural training programme for teen mothers, providing financial and material support for poor school children and contributing financially to cover the health insurance fee for the poorest households. Two local schools were visited: one private school and one public school. The tuition fee for the private school is 300 000 Rwandan francs (around 205 USD) per semester, while the public school is officially free but parents are expected to contribute 20 000 RWF (around 14 USD) per semester for the school meals. Though this fee is formally voluntary, some of the children from families unable to pay feel ashamed, which in some cases leads to school dropout. The public school accommodates approximately 3 000 students and to have space for all students, the school day is divided into two shifts: one morning to early afternoon, and one from mid afternoon to evening. Despite this arrangement, the observed classrooms were still overcrowded and the class sizes often reached around 60 students. This problem was also observed during public school visits in the quantitative data collection phase. Another problem for the poorest people in the village was affording public insurance, according to the founder, approximately 20% of the families couldn't afford the yearly insurance fee.

3.3.2 Bamboo plantation and workshop

The bamboo plantation visited was one of ARECO Rwanda Nziza's projects. The main purpose of the visit was to assess the availability of bamboo suitable for toothbrush production in Rwanda. The plantation is located in a buffer zone between Volcanoes National Park and adjacent rural communities and was established to encourage the villagers to cultivate bamboo and to decrease illegal harvesting inside the park by providing an accessible source of bamboo and medicinal plants.

The plantation is run by a women cooperative, and only few men are allowed to work there. During the visit, guidance was provided by a bamboo specialist and one of the few male workers on site. He introduced several bamboo species cultivated on the plantation. The bamboo that seemed the most suitable for toothbrush production was *Bambusa Arundinaria Alpina*, a native breed which he assured there is good availability of. After visiting the plantation, a visit to a local

bamboo workshop was made. In the workshop, lamp shades, baskets, vases and bamboo furniture such as chairs and coffee tables were produced. The lamp shades, baskets and vases were hand-woven by women, and the furniture was built by men who also operated the machines. The production environment was relatively simple, equipped with only basic machinery like a bamboo splitting machine and a table-mounted circular saw. Figure 4 shows photos of the local bamboo plantation and the workshop.



Figure 4. Unharvested Arundinaria Alpina, harvested and cut Arundinaria Alpina, splitting machine at one of the ARECO workshops, handicrafts made at the same workshop

3.4 Questionnaires

Quantitative data was collected through questionnaires handed out in two different schools. The objective of the questionnaires was to understand toothbrushing habits and toothbrush preferences of school children. The questionnaire was initially written in English and translated to Kinyarwanda before the study was initiated in the schools, with the help of retired professor Anne-Marie Uwitonze. The English questionnaire is available in appendix B, and the translated version in appendix C. The questionnaire was also reviewed by the project supervisor Julienne Murererehe and Anne-Marie Uwitonze. Before the children filled in the questionnaires, they were asked to read and sign an assent form that was written in Kinyarwanda, which was also initially written in English and then translated to Kinyarwanda by Anne-Marie Uwitonze. The same procedure was done for consent forms. The consent forms were signed by teachers, who signed their name and the name of the participating students. The English and Kinyarwanda versions of the assent and consent forms are available in appendix D and E.

The two schools were chosen by Anne-Marie Uwitonze, one of which being a public school (Rwankuba) and the other being a private school (Ecole Source Du Savoir Kabuga). Public schools are free of tuition fees in Rwanda [77], with the exception of school uniforms and school material. Private schools vary greatly, with some having a fee of around 300 000 (approximately 200 USD) per year [78], whereas higher-end international schools cost 2 250 000 RWF (approximately 1 500 USD) per year [79] and a more representative/average tuition fee is around 450 000 RWF (approximately 300 USD) [80]. The distribution of students in primary school in Rwanda who attend public and private school is around 93.5% and 6.5% respectively [81]. The data collected from the public school is thus more representative of the general population in Rwanda. The choice to include both one public school and one private school was made to understand the differences between these, as families with children who attend private schools are likely more financially secure than the families with children who attend public schools. Both schools were situated in the outskirts of Kigali. During the school visits when questionnaires were handed out, Anne-Marie Uwitonze assisted in explaining the project's purpose and the procedure of answering the questionnaire as this was explained in Kinyarwanda.

3.4.1 Method

A consecutive sampling strategy was used to get the sample size among P4 to P6 children in each school. The Sample size for the quantitative study is calculated with Yamane's formula:

n = Sample size
N = Population size
e = margin of error

$$n = \frac{N}{(1+N(e^2))} \tag{1}$$

$$n = \frac{480}{(1+480\cdot(0.05^2))} = 218 \tag{2}$$

With a population size of 480 (the estimated number of p4-p6 students in the selected schools) and a margin of error of 5%, the calculated sample size is 218 school children. To ensure that sufficient questionnaires were filled in, 242 questionnaires were handed out so that if some were lost or not handed back in, the study would still be able to reach the required 218 answered questionnaires. The questionnaire was answered in classrooms by 40 children at a time, where they were monitored by project members to avoid children speaking among themselves while answering the questions (with one exception where there were 42 students in one classroom).

After a recommendation from Anne-Marie Uwitonze, a question of whether or not the children had seen a wooden or bamboo toothbrush was included in the questionnaire to better understand the context of the question "which toothbrush would you prefer?"(regarding toothbrush material). She highlighted that since bamboo and wooden toothbrushes aren't available in Rwanda, it was unlikely that any of the children would have seen one before, and thus would answer "plastic" by default. Bamboo toothbrushes were also brought to the schools where the quantitative data was collected in order to show them to the school children to make their answer regarding preference of material more informed.

3.4.2 Result of questionnaires

The questionnaires were filled in by 242 students in total, 120 from the private school and 122 from the public school, as shown in Figure 5.

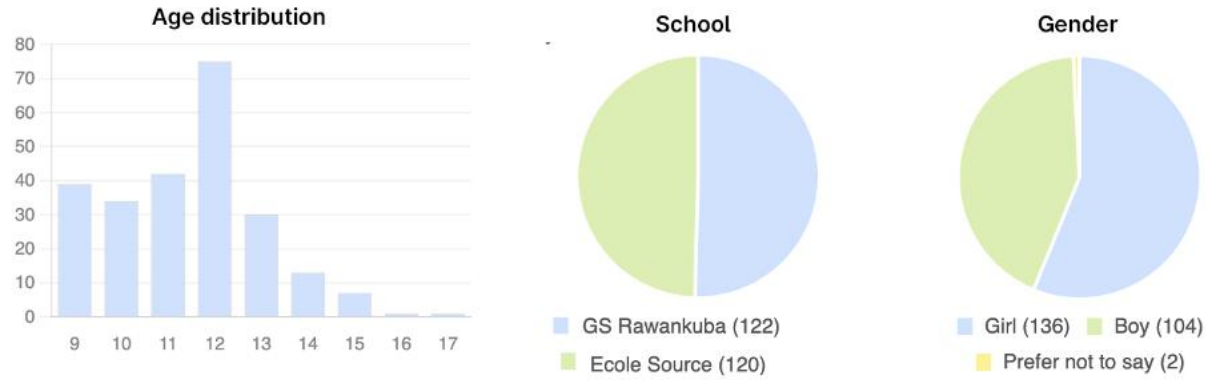


Figure 5. Demographics

All students in the private school replied that they have their own toothbrush while 19% of the students at GS Rawankuba replied that they didn't own a toothbrush. 11% of the respondents in GS Rawankuba changed their answers from "no" to "yes". This change could reflect social pressure, a desire to give the "correct" answer, or feelings of shame. Since similar changes were not observed to the same extent in other questions, the explanation that the changes were simply due to typographical errors was considered unlikely. Another possible explanation is that students sharing a toothbrush with family members, or who use wooden sticks or charcoal instead of conventional toothbrushes, were confused about how to reply to the question. Nevertheless, the gap between the private and public schools in access to toothbrushes is significant, as seen in Figure 6.

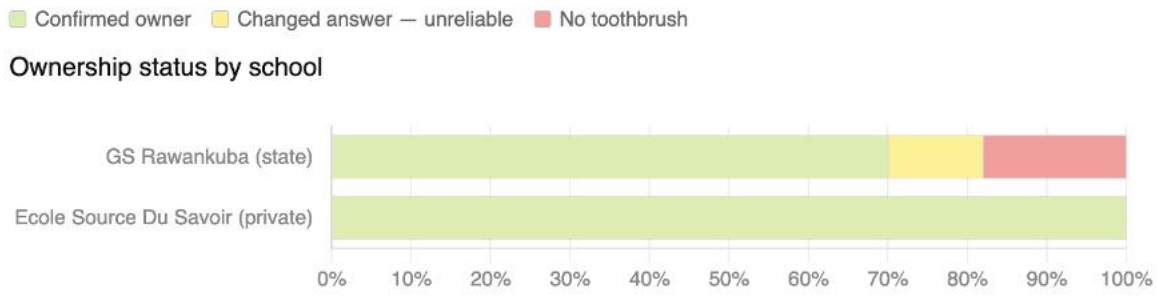


Figure 6. Ownership status by school

An unexpected finding was that the children in the public school to a larger extent use children's toothbrushes, see Figure 7. Possible explanations to this could be the price of children's brushes being lower than adult brushes, or an effect of less frequent toothbrush replacement. The majority of the toothbrush owners store their toothbrush standing up, in a glass or cup.

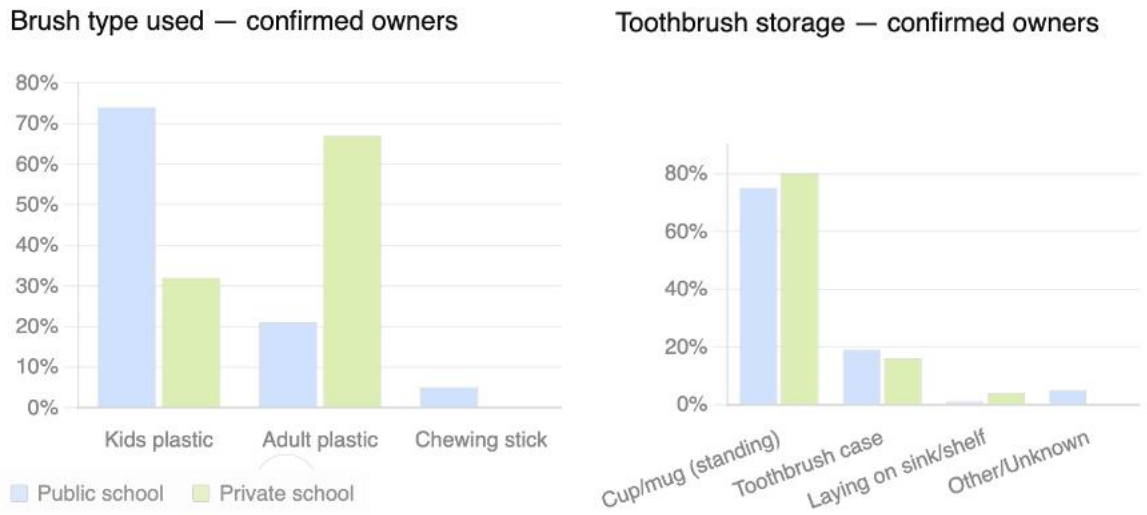


Figure 7. Brush type owned and toothbrush storage

Out of all participants, 53% felt pain in their gums when brushing their teeth. This could indicate either too hard-bristled toothbrushes, a need for guidance in toothbrushing technique or already inflamed gums due to poor oral hygiene. 27% of the children in the public school, and 21% of the children in the private school, had problems reaching all their teeth sometimes or every time they brush their teeth, see Figure 8.

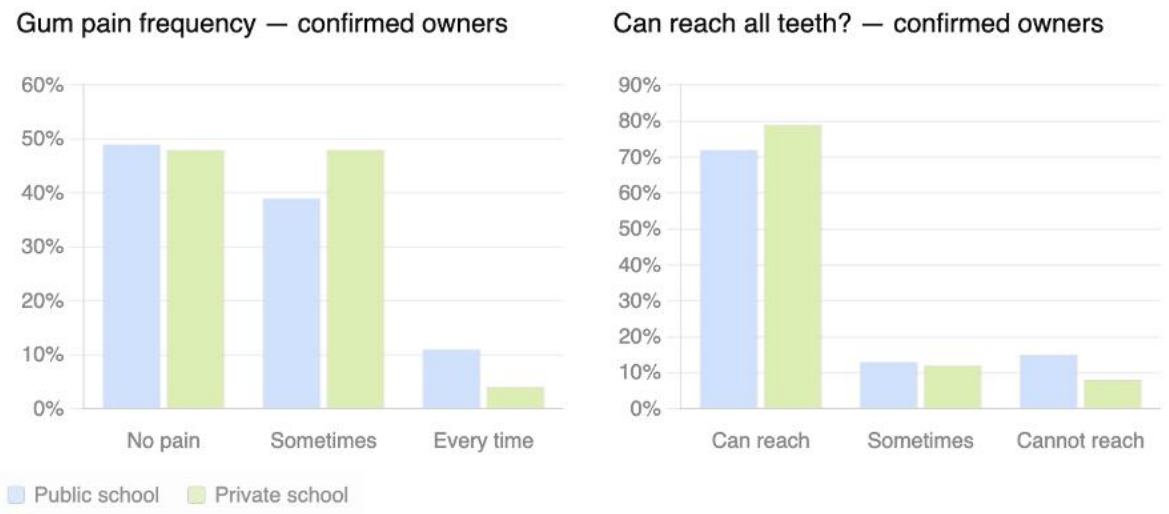


Figure 8. Gum pain frequency and reach of teeth

The question regarding whether the participants preferred bamboo or plastic toothbrushes was added to the questionnaire to assess the general acceptance of bamboo toothbrushes among the school children. This was of interest as bamboo or wooden toothbrushes are not currently available in Rwanda, and unfamiliar products may face acceptance hurdles. After being shown the sample of a bamboo toothbrush, the majority of the participants still preferred the plastic alternative. Children in the public school were more inclined to choose bamboo, but plastic remained the most preferred option overall, see Figure 9. When being asked what would make toothbrushing more fun, almost half of the children replied that a colourful toothbrush would make them more motivated. Personalization was also a popular option, 39% of the students in the public school and 32% of the students in the private school replied that having a toothbrush with their name on it would make toothbrushing more fun, see Figure 9. Additionally, in the free text alternative, 11 children (8 from the public school and 3 from the private school) wrote that having their own toothbrush would make toothbrushing more fun, which indicates that they currently might be sharing their toothbrush with family members.

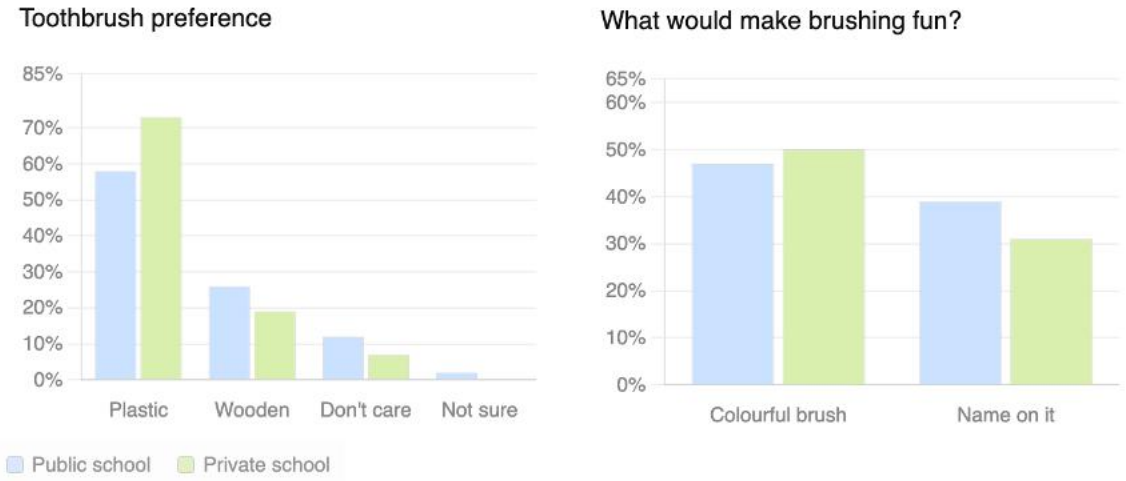


Figure 9. Toothbrush preference and motivation

4. Analysis



In this chapter, the findings from the literature study and field work are analysed and translated into design-relevant conclusions. This chapter represents the final part of the first diamond within the adapted Double Diamond process, where the collected information is structured, interpreted and narrowed down before moving into ideation and concept development.

The chapter summarises the analysis conducted by the authors, beginning with the main takeaways from the field work, before converging these insights into material selections and system-level considerations. The stakeholder map and stakeholder value network describe the proposed intervention system and should therefore be read as a proposal for how the system could function if implemented, based on the stakeholders and relationships identified during the project.

4.1 Main takeaways from field work

After the field work, the results were reviewed and summarized. Identified key insights were collected and formulated as main takeaways to guide the following ideation phase and to better understand the context in which the toothbrush is designed for.

4.1.1 Main takeaways from interviews

1. Bamboo is available and actively promoted, with existing plantations and institutional support for expansion.

2. Dendrocalamus bamboo is identified as a suitable species for manufacturing toothbrush handles, and is cultivated in the area around Kigali.

3. Woods like pine, eucalyptus, and cypress are already used in small-scale production and may be a viable option for toothbrush production.

4. Bioplastic toothbrush production is technically feasible, with potential for local production of granules within two years.

5. Industrial collaboration opportunities exist, and relevant contacts within bamboo and bioplastics manufacturing have been collected. ARECO Rwanda Nziza is a potential collaboration partner if bamboo is selected as material for the toothbrush, and Arth Biobag is a candidate if compostable bioplastic is chosen.

6. Waste management and recycling are reported as relatively functional by authorities, in contrast to the earlier literature findings.

7. Oral health interventions exist but are very limited in scale, reaching only a small number of schools. This indicates that there is already a willingness to improve the current situation.

8. Awareness is a key barrier to the lack of healthy oral habits in rural communities.

4.1.2 Main takeaways from the site visits

1. Some of the households can not afford small costs like health insurance and school meals, and being unable to pay even a voluntary fee causes shame and can lead to decreased attendance or in the worst case, dropouts. There should be no fees, voluntary or not, associated with the oral health school program.

2. Many public schools have overcrowded classrooms. The program should be possible to implement in a context where space and resources are limited. The intervention should be simple and time efficient, and easy to implement in large classes.

3. There is local bamboo suitable for toothbrush production. Bambusa Arundinaria Alpina is one of the suitable species that is thick enough and has a high availability. Women's cooperations in Musanze are growing this specific kind, which gives the opportunity to contribute to social impact by choosing them as a supplier. However, transportation costs to Kigali might be higher since Musanze is close to the northern border of the country.

4. Workshops have limited machinery and potentially limited electricity supply as well. For a large scale production, investment will have to be put in specialised machinery and training of staff.

4.1.3 Main takeaways from questionnaires

1. Access gap in public schools. All participating students in the private school stated that they have their own toothbrush while up to 30% of the public school children might not have a toothbrush. Considering that only a small percentage of the Rwandan children are private school students, the access gap seen in the public school is likely present across the country. Moreover, both schools were located in the capital Kigali, earlier studies have shown significant gaps in toothbrush ownership between the urban and rural population.

2. Bamboo toothbrushes are unknown and plastic is generally preferred. However, almost 30% preferred bamboo despite plastic being the norm. This may indicate a small but manageable acceptance barrier for introducing bamboo toothbrushes.

3. Many children experience pain in the gums when brushing their teeth. Soft bristles and guidance to good toothbrushing technique is a priority.

4. Personalisation and colour drives motivation. Especially in public schools where toothbrushes are more commonly shared between family members, having a personalised toothbrush is both a motivation and a hygiene factor. In school based oral health programs where in-school toothbrushing is one of the activities, personalised toothbrushes also serve a hygiene purpose.

5. Most children store their toothbrush upright in a glass or cup. The risk of water, toothpaste and saliva accumulating in the bottom of the cup should be considered. Design choices enabling a hygienic resting position, or minimising contact with brushes sharing the same cup, should also be prioritised.

4.2 Material selection

Building on information gathered during the desktop research and insights generated from the site visits and interviews, materials were evaluated by the authors and finally selected for the final toothbrush design.

4.2.1 Handle

Biodegradable plastic was discarded mainly based on complexity of production and lack of local availability. The required initial investment in machinery is big - injection molding equipment and molds are suitable for large-scale production but may not be feasible for starting small and scaling up the production gradually. Moreover, home compostable plastic toothbrushes are new on the global market which means there are questions around performance, durability and how these materials actually behave when they degrade. According to the interview with REMA, though the material is approved for single-use food containers in the country, home compostable plastic toothbrushes would have to go through testing to investigate how the material reacts to toothpaste, saliva and in a continuous use case. Another issue with biodegradable plastic is that it might get confused with conventional plastic and hence would be disposed of in regular waste and then end up either in a landfill where it won't decompose, or be sorted and recycled as plastic which would contaminate the other recycled plastic [82]. Although there is plenty of organic waste in Rwanda, the plastic granules available through the potential supplier Arth Biobag are produced in India. There are plans to establish local production of these materials from organic waste in Rwanda within the next two years, however, this development cannot be guaranteed and choosing to go for compostable bioplastic would thereby risk to compromise one of the key requirements of the toothbrushes.

Wood has similar qualities as bamboo when it comes to material properties and manufacturing opportunities, but was discarded because it lacks the government backing bamboo has. Since the government is a key stakeholder and the primary customer for the toothbrushes, choosing a material that aligns with national priorities is a strong argument. With the government supporting bamboo production, choosing bamboo could further motivate the implementation of a national-wide distribution of the toothbrush. According to the interview with Rwanda's wood value chain, bamboo is generally comparable or lower in cost than wood

Bamboo, specifically the native species Arundinaria Alpina, was chosen by the authors as the material for the toothbrush handle. It fulfills the requirements of local availability, biodegradability and the bamboo culms have dimensions appropriate for toothbrush production. Bamboo has several additional advantages that strengthen the decision: it is low-cost, grows quickly, biodegrades fully, and the government is investing in the bamboo industry. Bamboo is not currently used for large scale production, there are however bamboo workshops that work with bamboo and have some experience and machines applicable to the manufacturing of bamboo toothbrushes. Making use of already existing competence and machines by partnering up with one or more of these workshops would ease the initial stages of setting up production. Despite being focused on handicrafts and furniture, a collaboration would likely also decrease the time required for the workers to learn the new processing steps. Initial investments in a bamboo manufacturing line would be significantly smaller than for plastic manufacturing.

According to interviews with ARECO Rwanda Nziza, supply is not a concern in the long run since as long as the demand for bamboo is growing, the plantations will be expanded. Sourcing from local cooperatives would create jobs, encourage local entrepreneurship, and support environment protection activities such as planting bamboo in buffer zones around national parks

to protect the habitat of the mountain gorillas, and prevent soil erosion. By using bamboo, the project could create value not only for the end user, but also for local communities and ecosystems.

4.2.2 Bristles

Having chosen bamboo as the handle material, the bristle material was chosen. Boar hair and natural fibres are biodegradable and completely organic, making them a strong candidate for the project. However, natural fibres such as bamboo fibres, ridge gourd and coconut can not match nylon when it comes to plaque removal. Chewing sticks or miswaks are effective when using the right technique but require significantly more maintenance than conventional toothbrushes and are not widespread in Rwanda. The risk of a cultural stigma when using animal hair was considered too high, and the porous structure of boar hair absorbs water and potentially creates conditions for bacterial growth which would be a hygiene risk.

After ruling out the purely natural alternatives, Castor oil-based nylon was chosen by the authors as the most appropriate material. It has almost identical properties as conventional petroleum-based nylon, which has been dominating the market since the 1930s. Studies have confirmed that it can be produced in compliance with current ISO standards for toothbrush bristles, with plaque removal efficacy equivalent to conventional alternatives. The important difference between castor oil nylon and conventional nylon is that it is derived from a renewable resource instead of fossil fuels, which gives it a lower carbon footprint than standard nylon.

The drawbacks of castor oil nylon bristles are that it is non-biodegradable and not recyclable in the context of Rwanda. This means that the design will have to allow for separation of bristles and toothbrush body for proper disposal at the end of life since the toothbrush body will be biodegradable and the bristles won't. The bristles will be imported since there are no local suppliers in Rwanda, and setting up a castor oil nylon factory would not be feasible within the scope of the project.

4.2.3 Coating

Though bamboo is relatively water resistant compared to other woods, a coating was added to the design proposal to improve the life length of the toothbrush, especially given Rwanda's humid climate. Wax is commonly used as coating by established toothbrush brands, for example The Humble Co. use a natural plant-based wax on their bamboo toothbrushes [83], and WooBamboo use soy wax [84]. In a Rwandan context, beeswax is the strongest candidate and was thus chosen for the coating by the authors.

Rwanda has an established apiculture industry, with favorable climate and an abundance of wild bee populations. The beekeeping sector has been identified as an underutilized economic opportunity and a potential way of supporting rural development in the country [85]. The sector also has governmental support, as the government has recognised beekeeping as a strategic priority in the Strategic Plan for the Transformation of Agriculture [86], for example because of its contributions to biodiversity and rural income. Beeswax is a food-safe, non-toxic and locally available coating that offers effective moisture resistance [87]. It can be applied without the need for specialised equipment, simply by warming and rubbing it into the bamboo surface. These

properties, together with the government support and the opportunity to strengthen the social impact of the toothbrush production, made beeswax a well-suited choice.

4.2.4 Packaging

Kraft paper was chosen by the authors as the material for the packaging, specifically kraft paper bags. Since the intention is not to sell the toothbrush in stores, displaying the toothbrush well in stores is not a priority. Plastic packaging was rejected as this project sets out to design an environmentally friendly toothbrush with little to no plastic. One key reason for kraft paper bags being more suitable than cardboard boxes is that kraft paper bags are generally cheaper as less material is needed and the step of corrugation is not required. Additionally, cardboard boxes are heavier and less space efficient when transporting the toothbrushes. The main benefit of plastic and cardboard packaging is their ability to display the toothbrush, but as this factor is not important in this project, this benefit does not affect the choice of material in a significant way.

4.3 Stakeholder map

Based on the insights from the literature and user studies, 12 stakeholders were identified. To get an overview of their level of importance and how to manage them, they were mapped out on an interest-influence matrix, see Figure 10. This helped visualize which parts of the system to focus more resources on in decision-making and implementation, and who to simply monitor or keep informed. The stakeholder map was created by the authors as an analytical tool and does not describe an existing implemented system, but rather a proposal for how stakeholders would be involved in the oral health intervention.

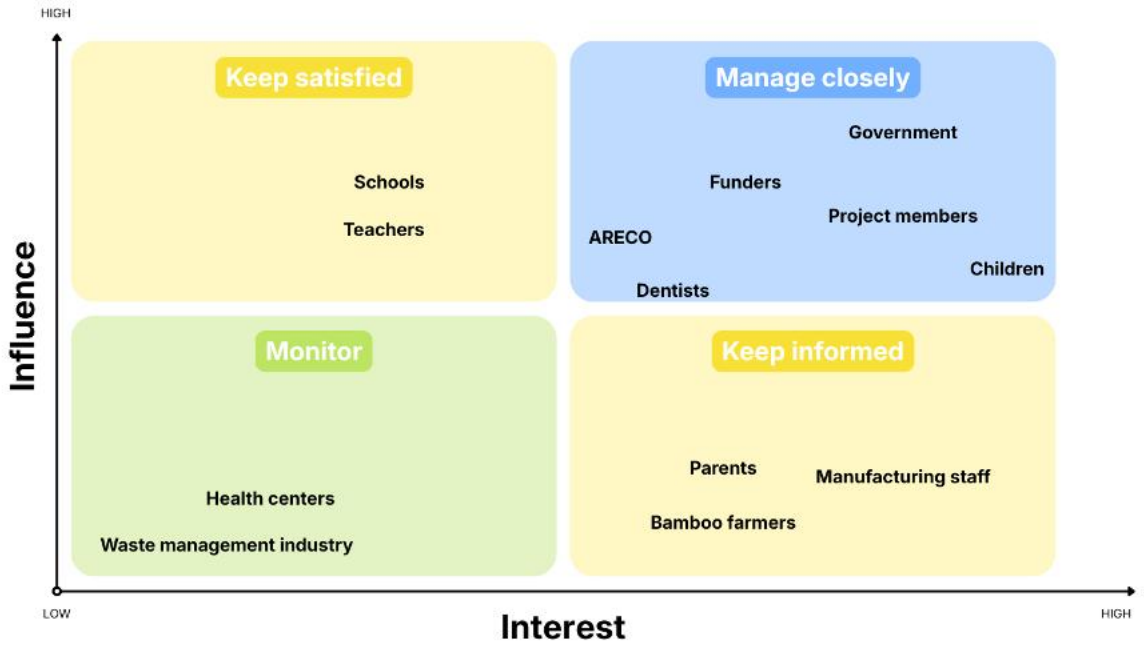


Figure 10. Stakeholder map

Government
The government plays a central role in the project, with both high influence and high interest. The school based oral health programme and the toothbrush production aligns with, and is affected by, several governmental strategies and policies such as the Made in Rwanda-policy, the plastic ban and the NOHSP. Since tooth and gum disease is one of the most common complaints among patients in hospitals, improved oral health across the population would lower the disease burden and the related costs significantly. As the objective of the project is to initiate a government-led and publicly financed oral health program, the Rwandan government will also act as the primary customer of the toothbrushes.

Funders
Funders carry both high influence and high interest in the project. Support by funders will be crucial for initial investments in machines and staff, to get the project running. In return, it is in the interest of the funders that the project generates measurable results like actual impact on oral health, fulfilment of SDG's or scalability. Currently, funding is still being secured, so the exact actors are not yet determined.

Project team
The project members from KTH, University of Rwanda and Scandinavian Dental Care are responsible for the project and for managing the stakeholders. They drive the project forward and have both a large influence and interest in the project.

ARECO
ARECO Rwanda Nziza has an important role in the project, acting as the link between the project team and the bamboo suppliers and workshops. They provide valuable insights in the current bamboo value chain, established relationships to key actors, and deep knowledge about bamboo. In return, the product system supports their mission of strengthening the local bamboo industry. Moreover, the implementation of the school program would increase their credibility with the government and other stakeholders.

Schools
The schools make the distribution of toothbrushes and carrying out the activities within the program possible by providing the context the program will be implemented in. They do not directly benefit from the program, but their collaboration is necessary for the program to work.

Teachers
The teachers are responsible for transferring oral health information and habits to the children and to make sure that the curriculum of the school program is followed. Thus, they have a high influence on the project.

School children
School children are the target group of the project. Having access to oral health tools and education early on in life creates healthy habits that may last for the rest of their lives. The project's success is largely determined by how well the product is accepted among the school

children, and how well they adopt the activities within the program. The children have high interest and high influence on the project.

Parents
The parents will notice the effects of the program by having a healthier child that requires less medical attention and not having to purchase new toothbrushes every three months. The healthy habits of the children will also potentially spill over on the parents and other family members. How parents respond to the new behaviors, whether or not they encourage the child, may also have a significant impact on the success of the project. Thus, parents have high interest in the results. and some influence.

Manufacturing staff
The staff in the factory have a high interest in the project, but little influence. The project strengthens local communities and local industrial development as it creates jobs and provides technical competence, since technical training is included in the job.

Bamboo farmers
The bamboo farmers have a high interest in the project, but little influence. They provide the factory with bamboo of the correct diameter and get a steady income in return. By strengthening women's cooperatives involved in bamboo cultivation, the project can contribute to improved livelihoods and financial independence at the community level.

Dentists
The dentists have a moderate influence on the project. The dentists will contribute with expertise in preventive oral care, formulating guidelines and educational content. They will provide oral health education for the teachers, which then will be passed on to the children. Their interest lies in improving oral health across the population.

Waste management industry
The waste management industry has low interest and low influence on the project. However, it is the stakeholder that determines what happens at the end of the toothbrush's use cycle. It is up to the waste management industry to make sure to sort the handle as biological waste and the head as mixed waste.

Health Centres
Health centres are not involved in the operations of the projects, however, if the project is successful they will eventually see the positive effects. As tooth or gum disease is one of the most common complaints among patients, a successful project could result in fewer visits, lower treatment costs and shorter queues.

4.4 Suggested stakeholder value network

To further visualize how the different stakeholders interact with each other within the proposed system, exchanges of tangible and intangible value of the key stakeholders was placed in a stakeholder value network, presented in Figure 11. The stakeholder value network was based on the principles from value network analysis [16] and was developed to use as a foundation for the implementation roadmap.

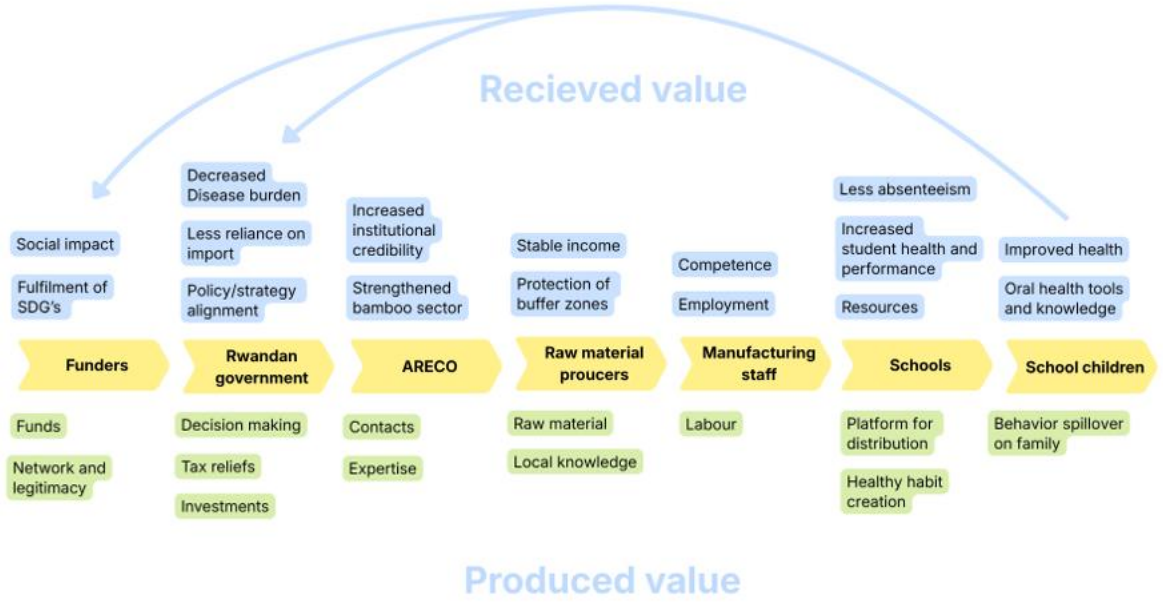


Figure 11. Stakeholder value network

The proposed value stakeholder network shows the relationships between the key actors involved in the proposed oral health intervention, with funders on one end and school children on the other. The network is illustrated as a chain since much of the value exchanges are structured in that way, the figure does however also show the interconnected relationships through arrows. Each stakeholder contributes value vital for the project's success, but also receives value in different forms from their contribution to the project. Funders bring capital and legitimacy, the Rwandan government provides tax reliefs, support and funds in return for a decreased disease burden, less reliance on imported, plastic toothbrushes and a stronger bamboo industry, in line with national strategies. ARECO Rwanda Nziza provides contacts and expertise at the same time as they are strengthening the bamboo sector. Bamboo farmers provide good quality bamboo, and receive a stable income while buffer zones are protected and local female cooperatives are empowered. Manufacturing staff contribute with labour and gain employment and new competence when they go through the technical training to work in the factory. Schools act as a platform for learning and distribution, and see improvement in student health which in the long run leads to improved results and less absenteeism. Finally, the school children improve

their health and tools to create long lasting oral health habits and positive behavior spillover on family members.

5. Ideation



This chapter presents the activities in the initial phase of the "Concept Development"-diamond. Based on the insights generated from the desktop research, interviews, site visits and questionnaires, ideas were explored in two brainstorming sessions: one focusing on idea generation using Design for the Base of the Pyramid (BoP) strategies, and several explorative sketching sessions visualizing ideas that came up during the BoP ideation session through sketching.

5.1 BoP Brainstorming

To generate a large number of ideas, the BoP brainstorming method was used. Based on Design for the Base of Pyramid strategies, four themes were formulated: Performance and function, Community and context, Economy and value, Environmental sustainability. For each theme, strategies relevant to this project were compiled from literature, and finally design ideas were generated from each of these themes and strategies. Figure 12, 13, 14 and 15 shows the mindmaps generated during the brainstorming sessions.

Performance and function
BoP consumers prioritize utility and the ability of a product to perform reliably in resource-limited environments [88]. The generated mindmap can be seen in Figure 12. The strategies for performance and function are:

Design for hostile environments:
Ensure the product is effective even with limited or contaminated water and can withstand high temperatures or dusty conditions common in BoP settings.

"Deskilling" the usage:
Create a simple and intuitive interface (such as visual prompts or colour-coded indicators) that requires no specialized training or high literacy levels to use effectively.

Functionality Transformation:
Rather than just offering a "cheap" product, consider if the product can be repurposed after its life span to maximize resource use.

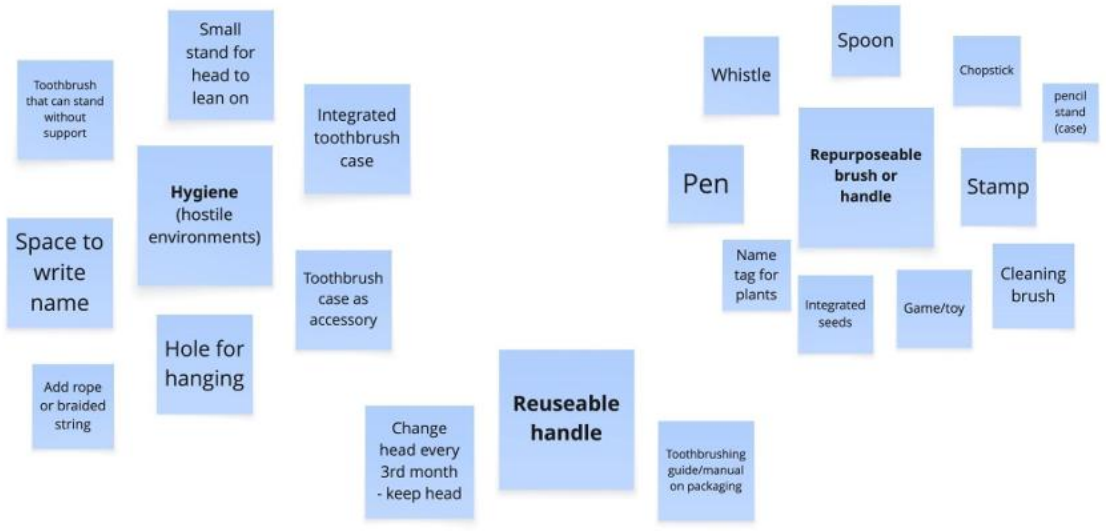


Figure 12. BoP mindmap - "performance and function"

Community and context
Embedding the product into the community context ensures acceptance and fosters long-term trust [88][89]. The generated mindmap can be seen in Figure 13. Strategies within community and context are:

Impact measurement:
Measure the product's success through its ability to improve holistic wellbeing (economic stability, health, and social status) across the whole community.

Multiple value creation:
Investigate whether the product or product system can be designed to create value across multiple sectors/communities.

Aspiration and Dignity:
Design the product as an attractive brand that provides status and choice, avoiding the perception of it being a "poor-person's product".

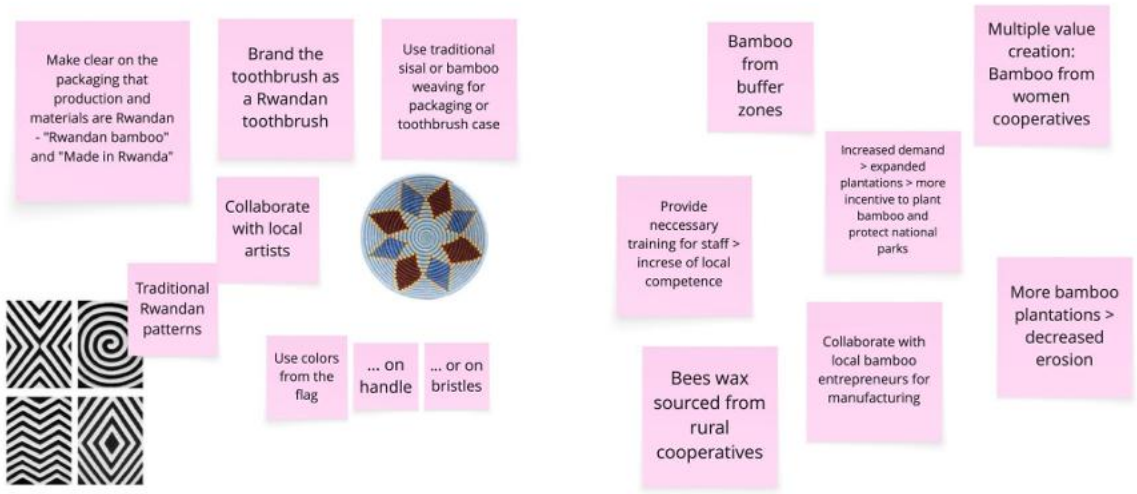


Figure 13. BoP mindmap - "community and context"

Economy and value
Finding ways to maximize value-for money is essential in BoP contexts. This can be done by for example minimizing prices by reducing manufacturing complexity, but also to provide products with higher value than a cheap, low-quality version [90][91][92]. The generated mindmap can be seen in Figure 14. Strategies for economy and value are:

Sachet Economics vs. Bulk Distribution: Use small unit packaging to lower the initial purchase barrier in shops, while utilizing larger bulk sizes for large scale distribution.

Scalability from Start:
Ensure the business model is simple and modular enough to be expanded.

Frugal Manufacturing:
Explore ways to reduce manufacturing costs by stripping away non-essential features and focusing on core performance.

Value over price:
BoP consumers are often more value-conscious than price-conscious, they need high performance for the price paid because they cannot afford products that fail.

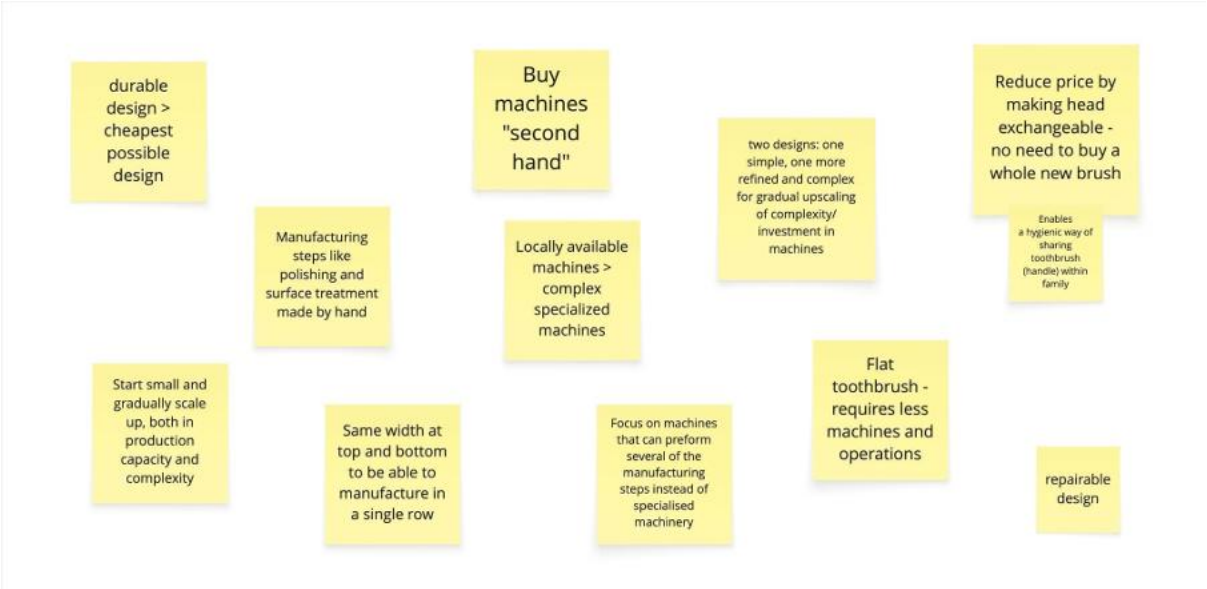


Figure 14. BoP mindmap - "economy and value"

Environmental sustainability
Designing in a BoP context, prioritizing the recycling of materials and a "doing more with less" mindset [91][92]. The generated mindmap can be seen in Figure 15. The strategies for environmental sustainability are:

Circular design and packaging:
Avoid plastic waste by utilizing biodegradable materials or refill systems that work within communities lacking formal waste collection

Local Material Sourcing:
Find ways to use locally available raw materials for production to reduce transport-related emissions and strengthen the local communities.

Waste Elimination:
Design the entire product life cycle with the goal that no waste should end up in nature, especially in environments where formal waste collection and recycling systems are absent

Long-term ecological impact:
Ensure that large-scale distribution does not lead to unintended environmental consequences, such as chemical runoff in local waterways.

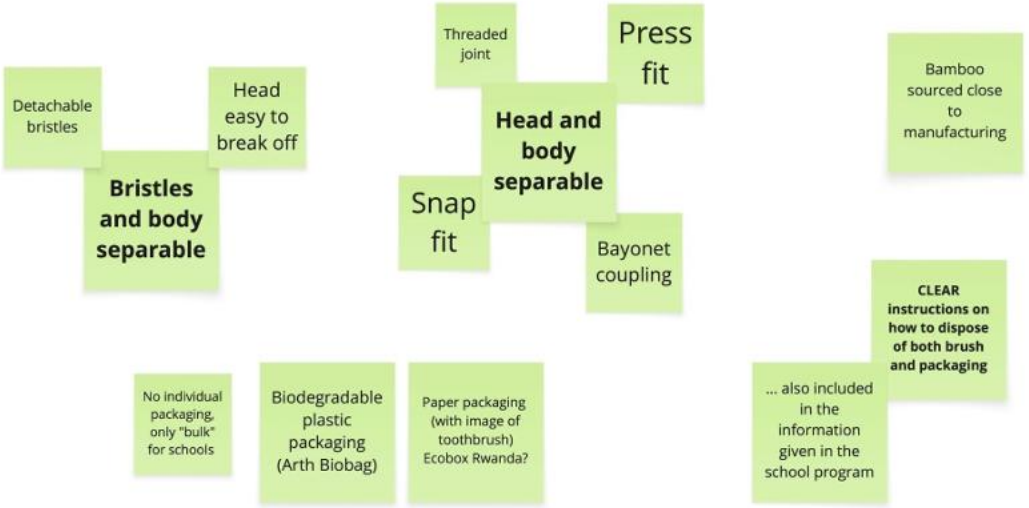


Figure 15. BoP mindmap - "environmental sustainability"

5.2 Explorative sketching

Based on ideas generated during the brainstorming session, explorative sketches were made to visualize ideas that regarded the actual toothbrush design. An overview of some of the sketches are presented in Figure 16.

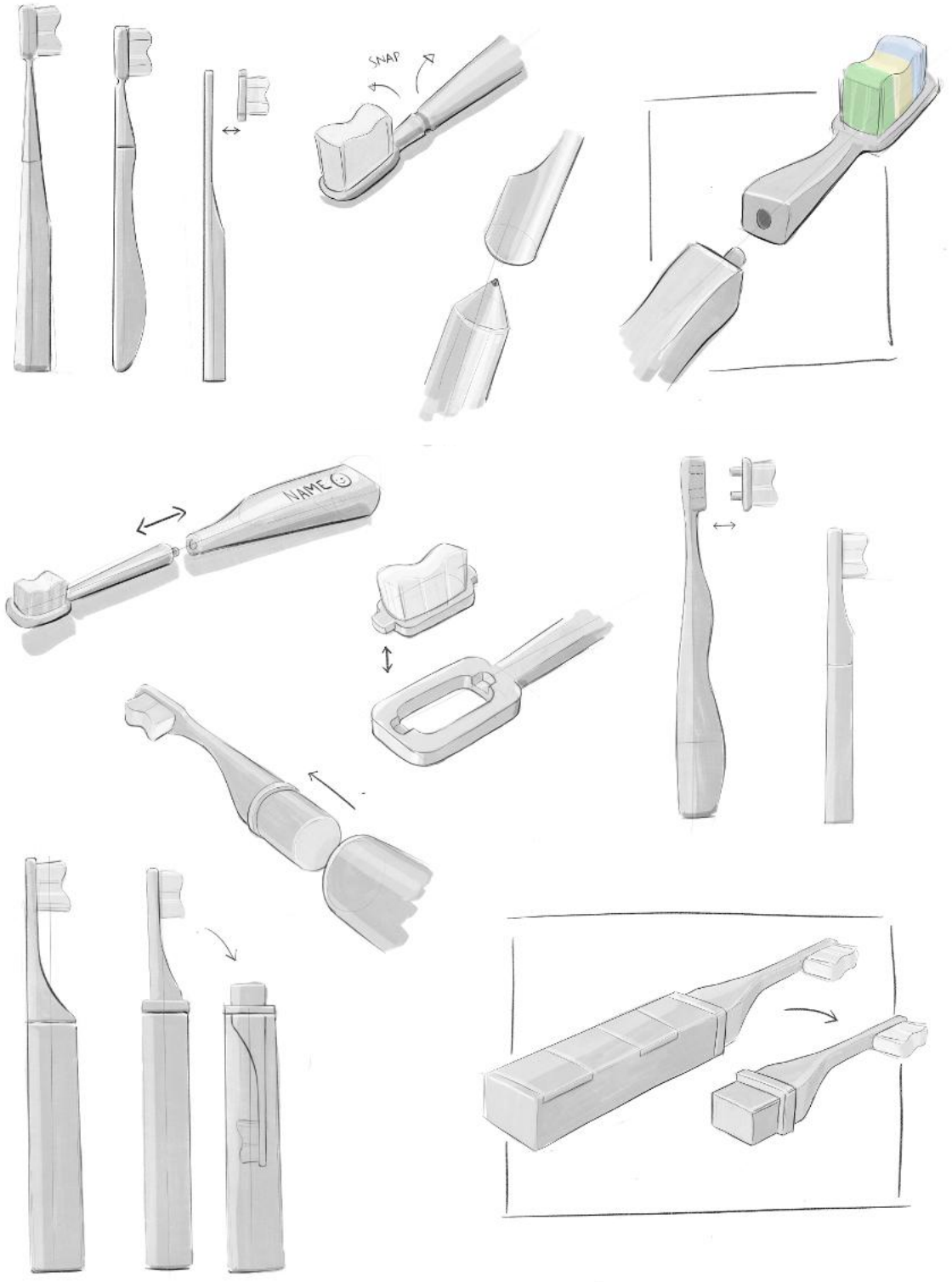


Figure 16. Ideation sketches

A recurring theme was ways of separating bristles and handle, both to enable separation of materials at disposal, and to extend the product's lifespan through allowing for reuse of the handle. Other ideas regarded allowing repurposing of the toothbrush at end of life by integrating a pen in the handle, having a designated surface for the user to write their names, and integrating a toothbrush case in the handle.

5.3 Concept selection

Following the explorative sketching, we evaluated the generated concepts against the criteria set in the project's aims and constraints: simplicity of manufacturing, local material compatibility, user suitability, and end-of-life considerations. Two concepts were selected for further development, each with a different approach to manufacturing complexity and investment.

The first concept prioritises simplicity of manufacturing, making it possible to produce using already existing workshops and equipment. All non-essential details are stripped away, resulting in a straightforward bamboo toothbrush with a comfortable handle, a slim head and wave-shaped bristles to improve access to all tooth surfaces. Instead of having a detachable head with couplings which would require additional manufacturing steps, the head is simply broken off the handle at end of life. Just below the toothbrush head, there is a groove in the handle, deep enough to allow the head to be snapped off by hand, but not so deep that it risks breaking off unintentionally during use. This allows the handle to be discarded as organic waste and the head as mixed waste.

The second concept requires more specialised machining and more specific requirements on the raw material, and was selected to explore what a higher-investment solution could look like. In this concept, the handle also acts as a protective case for hygienic storage and transport of the brush. The two concepts can be seen in Figure 17.

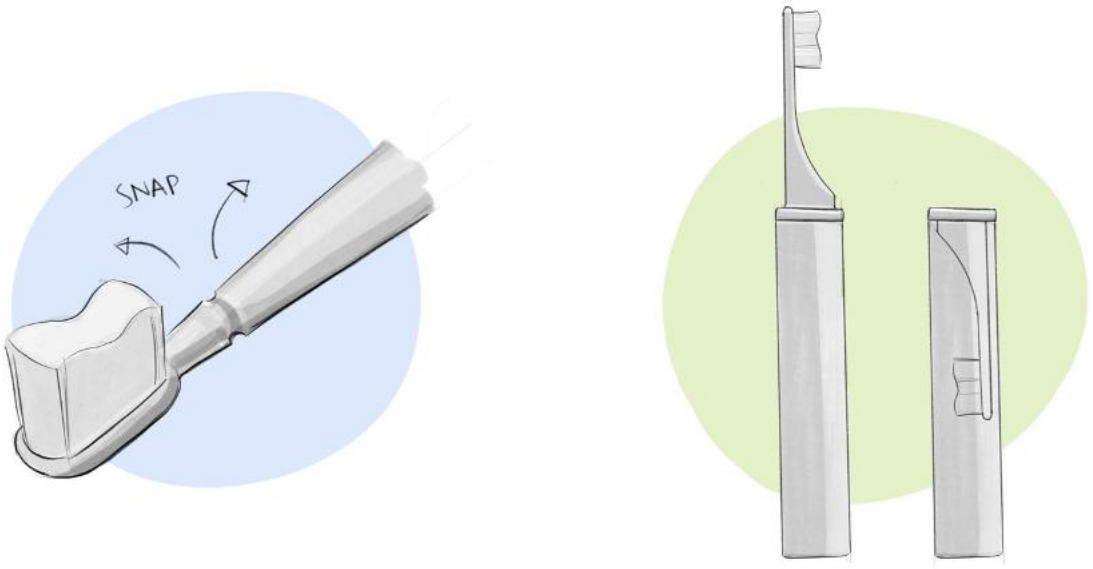


Figure 17. The two concepts.

The decision to develop two concepts was a strategy to manage risk and investment. The first concept allows for prototyping and proof of concept on a small scale without committing to specialised equipment and staff training. The second concept would then be introduced once the first has demonstrated its value and funding is in place.

6. Concept development



Still in the “development” phase in the concept development diamond, this chapter documents the process of refining the two concepts. First, important dimensions were decided based on existing toothbrushes. Then, the two designs were refined through 3D-modeling, sketching, prototyping and calculations.

6.1 Overall dimensions

To fit users of all ages, the design was divided into three sizes; 0-6, 6-12 and adult size. The measurements of each size was decided based on the measurements of 20 sample toothbrushes that were purchased for the project. The measurements of these toothbrushes can be found in Appendix F. The resulting measurements of the three toothbrush sizes can be found in Table 1.

Table 1: Dimensions of three different sizes of toothbrush

Size	Head length (mm)	Total length (mm)	Bristle length (mm)
Kids, 0-6 years	22	140	8
Kids, 6-12 years	26	160	8
12+ years	30	190	10

6.2 Development of concept 1

The development of Concept 1 focused on two areas. The first was finding a groove depth that would allow the brush head to be snapped off intentionally at end of life while being durable enough to withstand everyday use without breaking. The second was exploring different handle shapes to find a design that is comfortable to grip and is visually appealing to encourage use.

6.2.1 Manufacturing

The goal of Concept 1 was to keep manufacturing as simple as possible without compromising the toothbrush's core functionality. To limit the number of required operations and avoid the need for specialised equipment, the brush was given a two-dimensional profile to require milling in one direction only, while still comfortable to hold. The head and handle end were also designed with identical widths, so that multiple brushes could be routed in a continuous row and separated in a single cutting step.

6.2.2 Calculation of groove depth

To determine an appropriate groove depth, a bending moment analysis as well as prototype tests were performed. The purpose of the calculations was to estimate a maximum allowed groove depth to prevent unintentional breakage during use, while the purpose of the prototype tests was to find a groove depth that was comfortable to break intentionally. The calculations were made by creating a matlab script, which then could be used while iterating dimensions of the toothbrush. The script can be found in Appendix G. The brushing force Rt was set to 10 N, based on the maximum brushing force of 2 N [93] multiplied with a safety factor 5. A safety factor of 5 was chosen given the importance of durability in the BoP framework, and to account for uncertainties in materials and manufacturing. The force diagram is illustrated in Figure 18.

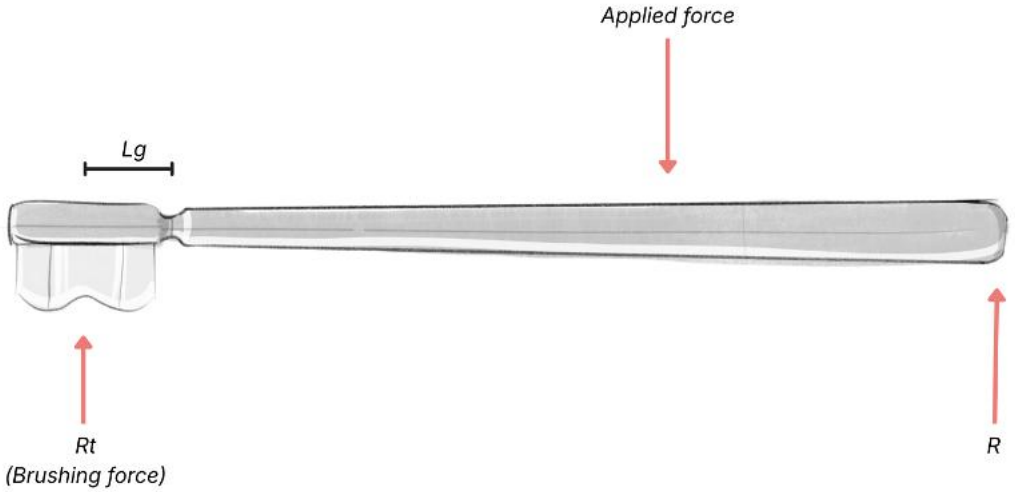


Figure 18. Force diagram

The brushing force was used to calculate the bending moment at the groove. The required cross-sectional height at the groove was then found using the bending stress formula in equation 4, with a stress concentration factor *Kt* of 1.5, based on standard stress concentration factors for rounded grooves and bending section modulus *W* calculated in equation 5. A bending strength σ_b of 80 MPa was used, which is the bending stress in the weakest direction, across the fibres, and therefore the most conservative assumption.

$$R_t = F \cdot Sf \tag{2}$$

$$M = R_t \cdot L_g \tag{3}$$

$$\sigma_b = K_t \cdot \frac{M}{W} \tag{4}$$

$$W = \frac{b \cdot (h-x)^2}{6} \tag{5}$$

From these equations, the groove depth can be isolated and calculated with equation 6.

$$x = h - \sqrt{\frac{K_t \cdot 6 \cdot M}{b \cdot \sigma_b}} \tag{6}$$

When inserting the dimensions of the adult toothbrush into the script (see Table x), with an original cross-section of 5 x 5 mm at the groove, the calculated groove depth was 1.6 mm. After calculating the upper depth limit, a prototype test was carried out to estimate a comfortable breaking force. Three simple toothbrush handle prototypes were made from Arundinaria Alpina bamboo, with the adult toothbrush dimensions and groove depths of 1 mm, 1.5 mm and 2 mm respectively, see Figure 19. Each prototype was then tested to assess the required breaking force.



Figure 19. Prototype tests

In line with the calculated results, the handle with a 2 mm groove broke easily while the handles with 1.5 mm and 1 mm grooves provided enough resistance to prevent breakage during use. Between the two safer options, the 1.5 mm groove snapped off without excessive effort and the 1 mm groove required a little more work. A groove depth of 1.2 mm was finally selected for the design as it provides both safety during use and ease of separation at end of life.

6.2.3 Shape variation

To find a handle shape that would be comfortable to hold and visually appealing, different shapes were modeled in Blender. Figure 20 shows an overview of the modeled shapes. After selecting a few favorite models, the selected models were 3D printed and tested. The dimensions were based on the decided measurements for the different toothbrush sizes, and the maximum width of the handles was constrained by the geometry of the split and flattened bamboo planks. Based on information from ARECO Rwanda Nziza, the indigenous species Arundinaria Alpina available in the north of Rwanda typically grows to around 8 cm in diameter with a wall thickness of about 1 cm . Split into eight sections and flattened, each plank yields a maximum of roughly 23 × 7 mm, which are the outer limits for the toothbrush body dimensions.

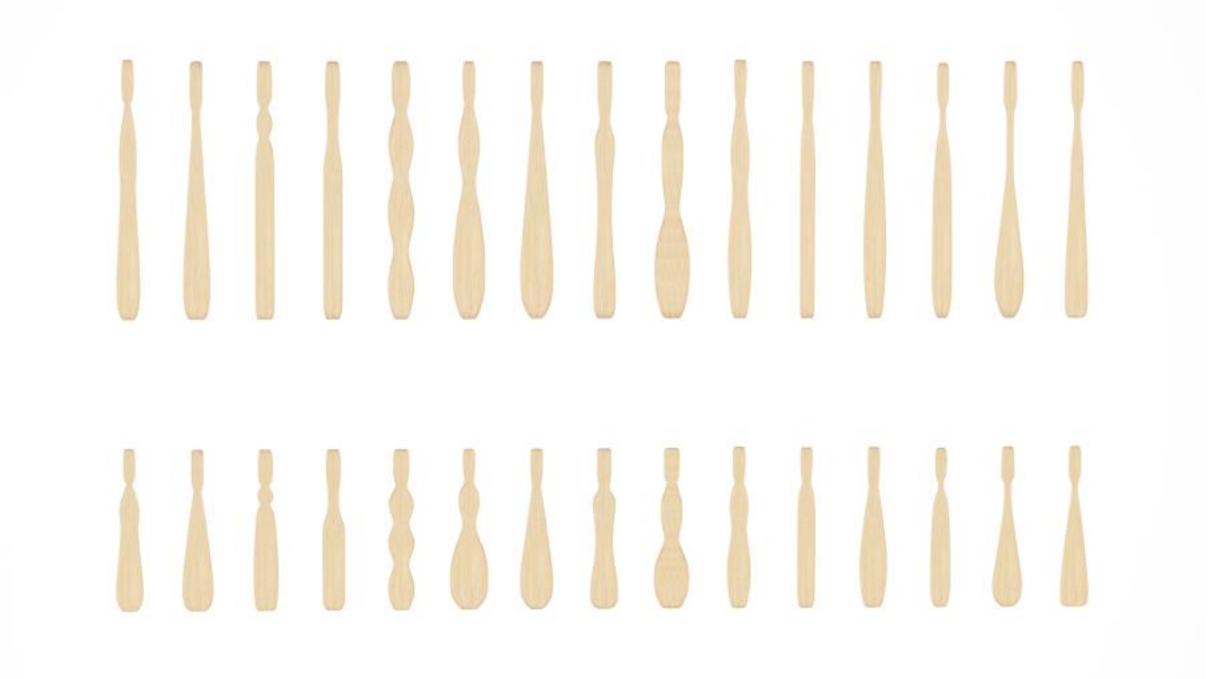


Figure 20. Shape variation of kids and adult models

Instead of making the toothbrush for the youngest children a smaller version of the adult toothbrush, it was designed to fit a child's hand. According to user observations made by IDEO, children tend to hold the toothbrush in their fist rather than gripping it with their fingers like adults commonly do [94]. By widening the handle, the design accommodates this natural fist-grip and provides a more comfortable brushing experience for young children.

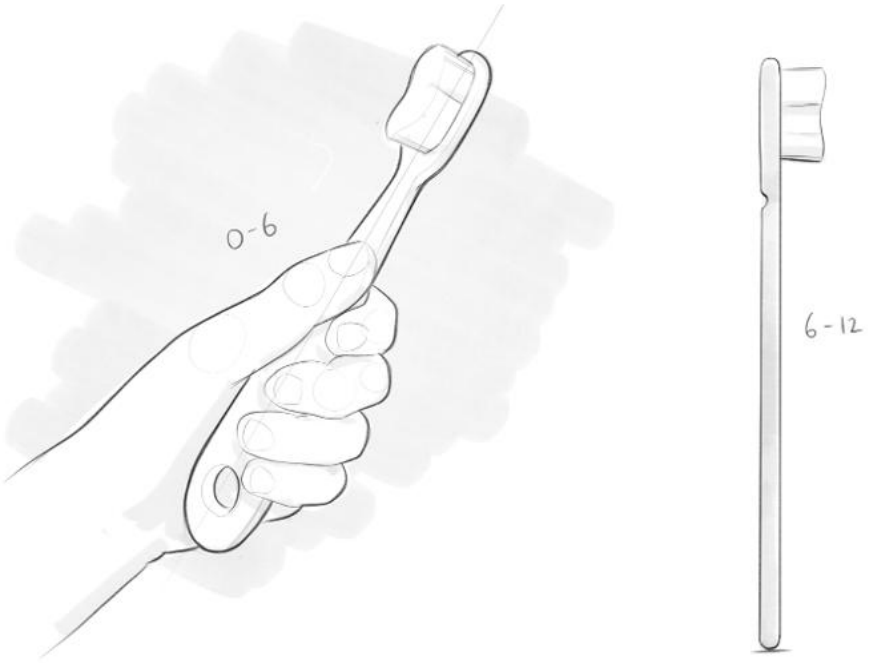


Figure 21. Sketches of toothbrush models for 0-6 and 6-12 years.

6.3 Development of concept 2

The development of concept 2 focused on three main areas: design alterations to make manufacturing easier, the explorations of different shapes and the tolerance for the press fit between the two pieces.

6.3.1 Design for manufacturing

Two different shapes were considered for concept 2, a handle with a circular and a handle with a rectangular cross section. Figure 22 shows sketches of the two variations.

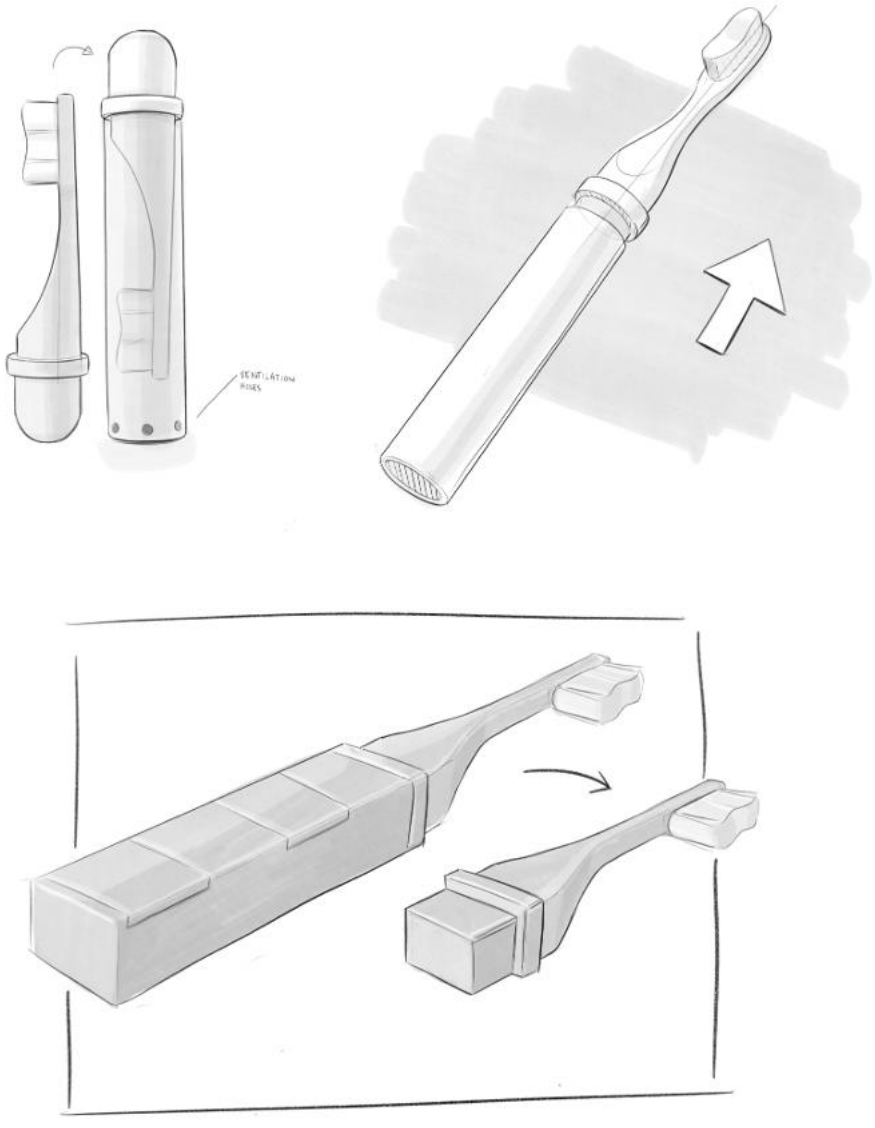


Figure 22. Two considered models for concept 2.

They would differ a lot in production, where the rectangular one would be constructed from two intersecting pieces glued together and the circular one would be constructed from a single

relatively small culm section. The circular option was chosen as the production of this would require less steps. The round shape also allows for a more comfortable grip. The limitation that this shape presents is the size of the bamboo used since the diameter of the inner wall cannot be larger than the hole of the handle where the toothbrush head is placed. Additionally the diameter of the outer wall must be larger than the outer diameter of the handle. This limitation would be overcome by sourcing bamboo that meet these requirements, either by using a bamboo species with these dimensions when mature or by harvesting a younger Arundinaria alpina bamboo. Making further use of bamboo's structural properties, the closed side of the handle is also made from the same piece of bamboo culm as the rest. This is possible since at each node of a bamboo culm there is a horizontal inner wall which would form the closed end of the handle, with the other end being open. Figure 23 shows a split Arundinaria Alpina culm where the node is visible.



Figure 23. Split arundinaria alpina bamboo culm.

To avoid many excessive manufacturing steps of the toothbrush head, this too is made from one piece of bamboo.. It has a protruding rim that reduces the amount of liquid entering the handle and to prevent the head from being inserted too deep into the handle. This rim would be created initially having a thicker dowel which would then be lathed to the diameter of the hole, except for the rim. For this process, a larger bamboo would be sourced, as the wall thickness of the bamboo would have to be larger than the diameter of the rim.

For the bottom part of concept 2, the handle doubling as a case, a cylindrical shape with one open end must be manufactured. As bamboo is cylindrical, the base would be manufactured from a piece of an unprocessed culm where the inner diameter is smaller than that of the inner diameter of the finished product, and the outer diameter of the unprocessed culm is larger than that of the outer diameter of the finished product. This process would make use of the cylindrical shape and only a lathe would be required to reach the desired handle shape. Bamboo culms are divided into segments at the nodes. At each node, there is a node plate. This means that the culm would be cut close to each node, and one end of the cut pieces is open and the other is closed by the node plate. For this manufacturing to be possible, a bamboo that has the correct inner and outer diameter and a suitable length between each node is required. Additionally, a hole would be drilled in the bottom. The hole would allow air and water drops to escape the case. The hole would allow for air to flow in only one direction at a time. To allow for better airflow, an additional hole was added so that air can flow in and out simultaneously. The

added hole should however not risk water, toothpaste and saliva to enter the handle while toothbrushing. This was solved by adding a hole that is only present when the toothbrush head is in the handle, and closed when the toothbrush is in use.

6.3.2 Shape variation

After choosing a cylindrical shape for the handle, different shapes of the upper part of the toothbrush were explored through drawing, 3D modeling and 3D printing. Considered shapes can be seen in Figure 24. The different shapes were then evaluated by ease of manufacturing, comfort and visual appearance.

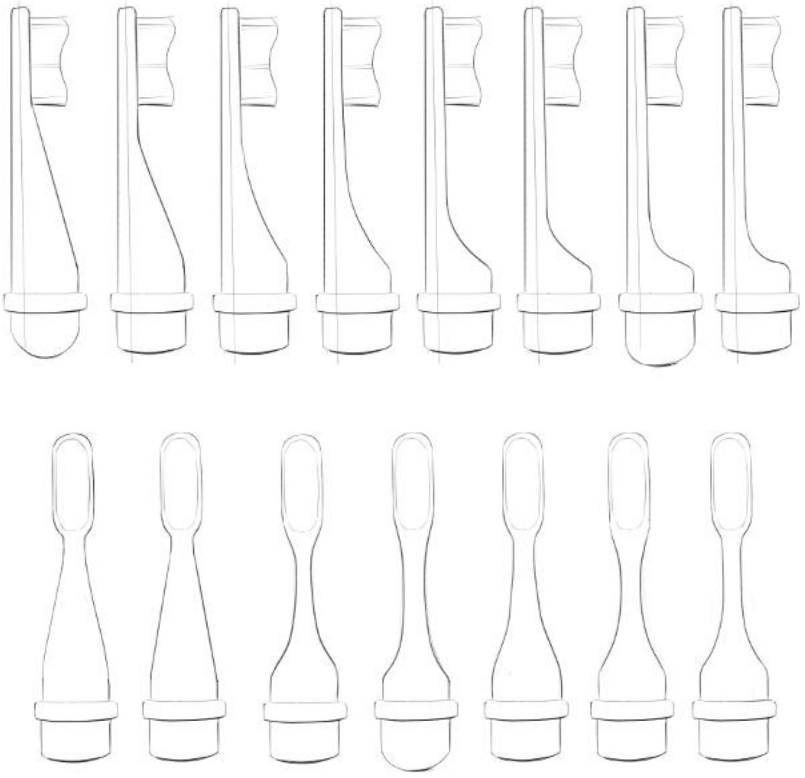


Figure 24. Shape variation of upper part of concept 2.

6.3.3 Press Fit Tolerance

The hollow cylinder (handle/case) and the solid cylinder (toothbrush head) will be connected with a press fit. For the press fit to offer sufficient friction during toothbrushing, yet not risking breaking either part when inserting or removing the toothbrush head, the exact margin would have to be a result of tests. The removal of the toothbrush head should also not require too much force, as a child should be able to perform this action. During tests, the factors that should be considered when deciding or clearance are insertion force, retention force and risk of outer cylinder cracking. The clearances tested should be positive as no tolerance or a negative tolerance would likely result in the bamboo cracking. As bamboo is a hygroscopic material, it can be expected that the connection points vary slightly in size over time and clearances around 0.1 mm

should be tested. The testing should also include the material being exposed to water, as the toothbrush will often be in contact with water.

7. Result



This chapter presents the “Result & Implementation” diamond. Through desktop research, site visits, interviews and questionnaires, followed by concept development, the results of this project include the design of two concepts and a suggested implementation roadmap. For the designs, much focus was put on the manufacturing capabilities in Rwanda as well as the context in which they will be used.

7.1 Concept 1

In this section, the design proposal for concept 1 is presented.

7.1.1 Design

Concept 1, seen in Figure 25, is a simple toothbrush design with a flat side profile. The handle is made of Arundinaria Alpina bamboo and the bristles are soft castor oil in a flat surfaced bristle configuration. It is designed to be manufactured with cheaper machines than those conventionally used for bamboo toothbrush manufacturing. The brush head has a slim shape to improve reach. The bristles are soft to prevent gum irritation and coloured blue, yellow and green, representing the Rwandan flag. Using the colours of the Rwandan flag ties back to the BoP "community and context"- strategies. By giving the toothbrush a cultural identity, the design is meant to instill a sense of pride and belonging in the user.



Figure 25. Overview of concept 1.

On the opposite side of the bristles, below the toothbrush head, a groove is located in order to more easily be able to break off the toothbrush head. This is done in order to separate the compostable material from the non-compostable material - the nylon from the bamboo. The groove also acts as a small reminder for the user that the head should be broken off at the end of life. The groove can be seen in Figure 26.



Figure 26. Close-up images of handle, groove, and bristles

Concept 1 comes in three different models, for ages 0-6, 6-12 and 12 and up. The toothbrushes for ages 0–6 and 6–12, seen in Figure 27, are wider than the 12+ version, providing a more comfortable grip for children's hands while also offering enough space for users to write their name. To increase the comfort when gripping the toothbrush, all of the three models have rounded edges. At the head, the round is smaller to leave more space for bristles to be inserted. The holes on the bottom of the toothbrushes for children of primary school ages, 0-12 years, let the users hang their toothbrushes on a hook, for drying after use and storage, see figure 27. As the toothbrushes are intended to be used in schools, this solution offers a more hygienic alternative to storing several toothbrushes in one cup, risking bristle-to-bristle contact. For full measurements of Concept 1, see Appendix H.



Figure 27. Toothbrush models to be used in schools (image generated with Leonardo.Ai, and refined with photoshop, prompts used for the image generation can be found in Appendix I).

7.1.2 Manufacturing

Concept 1 would be manufactured in Musanze, in current bamboo workshops close to where the bamboo is grown. The manufacturing process of concept 1 is adapted from a conventional bamboo toothbrush manufacturing process to enable smaller initial investments. Below in Figures 28, 29 and 30 are illustrations of the state of the bamboo for the entire production, from raw bamboo to finished toothbrush in its packaging.

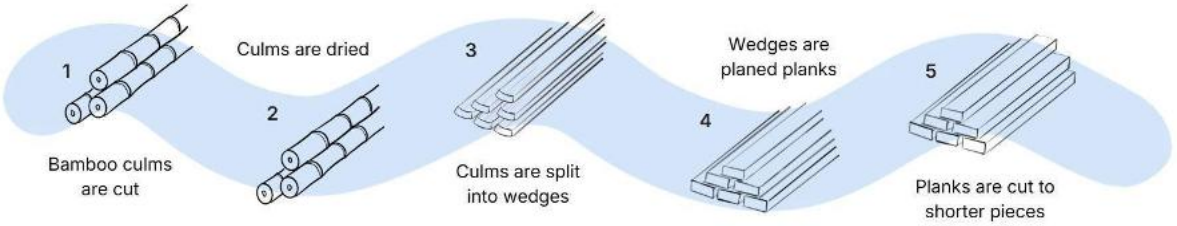


Figure 28. Manufacturing of concept 1, steps 1-5

1. Raw bamboo is cut with a sawing machine already existing in Rwanda.
2. The bamboo culms are dried.
3. The bamboo culms are split into wedges with a splitting machine that already exists in Rwanda.
4. The wedges are planed with a thickness planer into planks.
5. The planks are cut to shorter pieces with a sawing machine.

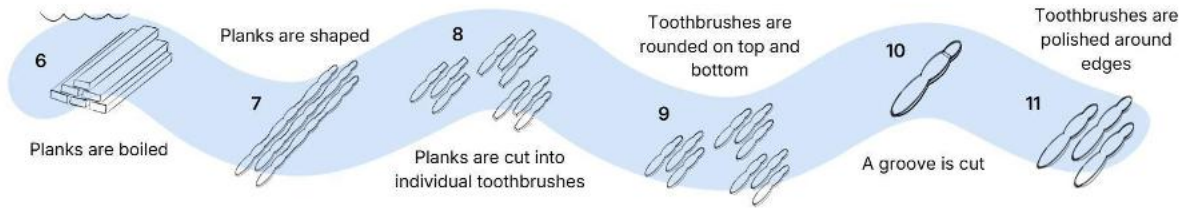


Figure 29. Manufacturing of concept 1, steps 6-11

6. The planks are boiled in water and dried to remove starch and sugar.
7. The planks are cut into the shape of the toothbrush on a router table with a template. The template consists of multiple toothbrushes, so each plank is cut into multiple linked toothbrushes.
8. The planks are cut into individual toothbrushes with a sawing machine.
9. Toothbrushes are rounded on the top and bottom with a sanding belt to remove the flat sides.
10. The groove on the back of the toothbrush is created by cutting with a router.
11. The toothbrushes are polished with sandpaper along the edges to remove sharp edges.

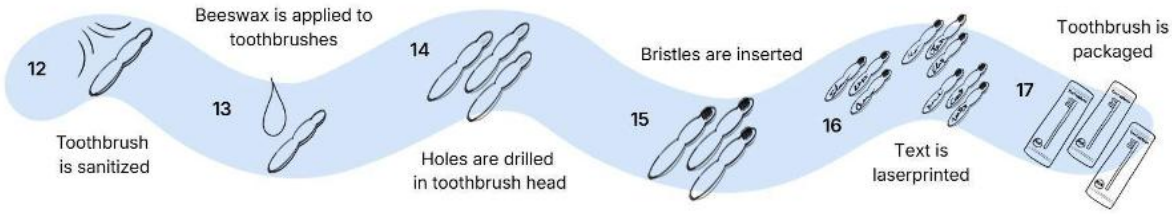


Figure 30. Manufacturing of concept 1, steps 12-17

12. Dust from shaping is removed and the toothbrush handles are then sanitized using hot water or steam. The products are then dried in a controlled environment.
13. A beeswax is applied as a protective coating. The coating is melted and applied with a cloth. The toothbrush is then left to cure for 24 hours at room temperature, to let the wax solidify.
14. Holes are drilled in the toothbrush heads.
15. Bristles are inserted in the drilled holes. Step 13 and 14 are completed in the same drilling and tufting machine.
16. Text is laser printed on the toothbrushes with a laser printer.
17. The toothbrush is packaged.

This process would use some machines that already exist in Rwanda, some machines that are much cheaper than the ones that are usually used, and some machines that are required for small and large scale manufacturing. The machines that would substitute standard toothbrush manufacturing machines are a thickness planer, a router table, a sanding belt and a plunge router. Machines that no longer are required are “Elementary bamboo strip planing machine”, “High speed bamboo strip further planing machine”, “Bamboo toothbrush milling machine”,

"Toothbrush milling machine", "Toothbrush handle polishing machine". The machines that are no longer required and the new machines that would be required and their prices are listed in Table 2.

Table 2: Prices for machines used for the two different manufacturing setups

Machines in conventional manufacturing		**Machines used in new process**	
Machine	**Price [USD]**	**Machine**	**Price [USD]**
MBXD-10 (Elementary bamboo strip planing machine)	9 980	Cocraft-HP317 (Thickness planer)	375
MBHB-4 (High speed bamboo strip further planing machine)	8 400	DeWalt DWE7492 (Router table)	586
BLW-400 (bamboo toothbrush milling machine)	10 200	Cocraft HBD500 (Belt sander)	150
RPG-5 (toothbrush handle polishing machine)	5 000	Bosch POF 1200 AE (Plunge router)	140
Total: 33 580 USD		**Total**: 1 251 USD	

For the machines stated in the standard manufacturing process, prices and specific machines were provided by the machine manufacturer Besta Bamboo Machine Co., Limited. The total price for the machines that are required for both manufacturing processes is 44 500 USD (for a drilling machine, tufting machine, precise cutting machine and a logo branding machine). The total price for machines of each setup would therefore be 78 080 USD and 45 751 USD, the difference being 32 329 USD. The vast majority of the machine cost for the new setup is comprised of the machines used for both setups, and those machines would be used when the production is scaled up and concept 2 is manufactured.

7.2 Concept 2

In this section, the final design proposal of concept 2 is presented.

7.2.1 Design

Concept 2 has a more complex design than concept 1 as it is intended to be produced and implemented to the school based oral health program when larger investments can be made. It consists of two pieces, the toothbrush head and the handle. The bristle configuration and material is the same as in concept 1, with soft bristles made from castor bean oil in a flat surface configuration. This concept also includes three different sizes to fit ages 0-6, 6-12 and 12+ that are shown in Figure 31.



Figure 31: Overview of the three different sizes of concept 2.

The handle is a hollow cylinder in which the head can be pressed into. The toothbrush head has a rim that prevents the head from being pushed in too far into the handle. Above and below the rim, the toothbrush head is cylindrical with a diameter of 18 mm, which matches the inner diameter of the handle. This enables the toothbrush head to be inserted with either the head up or down as seen in Figure 32. The rim also prevents water and toothpaste from entering the handle, acting as a cover. While the head is pointing down, it is encased by the handle so that the head will not come in contact with dirt. Making it easy to hygienically transport the toothbrush is important as the toothbrush will be transported to and from school often.



Figure 32: Toothbrush head in different positions.

It is only the toothbrush head that will be in contact with the user's mouth and will likely wear faster, additionally toothbrush heads should be replaced every three months as the bristles get worn down and lose effectiveness. When replacing the toothbrush, only the toothbrush head has to be replaced and the handle can be used for multiple toothbrush heads. The length of the toothbrush head is designed to reach the molars, with the wider cylindrical part still being outside the user's mouth. The toothbrush handle has a flat base and can thus stand on flat surfaces by its own, enabling storage outside of a shared cup. The handle is open where the toothbrush head is inserted, and on the other side it is closed to reduce dirt that enters the handle. For water to escape and air to flow through the handle, the closed bottom has a hole as seen in Figure 33. For the hole not to be covered when placed on a completely flat surface, the outer cylinder extends slightly. In the extending part there are also holes so the air can flow through. An additional hole, in the toothbrush head, allows for the air to flow in and out from the handle simultaneously. The hole starts from the rim of the toothbrush head and extends slightly longer than where the upper cylinder of the toothbrush head transitions to a flat surface, as shown in Figure 33. This allows for the hole to only be present when the toothbrush head is in the handle, and the seal between the toothbrush head and the handle remains when the toothbrush is in use. The diameter of the hollow cylinder in the handle is dimensioned around the toothbrush head. There is a small margin between the bristles and the handle when the toothbrush head is inserted in the handle, see Figure 33. The wall thickness of the handle is 2 mm. Figure 34 shows the toothbrush in the intended context. For full measurements of concept 2, see Appendix H.



Figure 33. Close-up of the upper hole. Bottom of the handle with holes for air flow. Toothbrush head being covered by the handle.



Figure 34. Toothbrush in context (image generated with Leonardo.Ai, and refined with photoshop, prompts used for the image generation can be found in Appendix I).

7.2.2 Manufacturing of concept 2

Concept 2 would be manufactured in Kigali special economic zone and the first processes of the bamboo manufacturing - drying and splitting, would be performed in Musanze where the bamboo is grown.The manufacturing of concept 2 is divided into two different processes, as it consists of two parts. The toothbrush head will be manufactured from a thick walled bamboo, as the thickness must be at least as large as the thickest point of the toothbrush head. The

toothbrush handle will be manufactured from a smaller bamboo, as the diameter of the natural hole (internode) must be smaller than the diameter of the hollow cylinder in the handle, while the outer diameter has to be larger than the outer diameter of the handle. The manufacturing steps for the toothbrush head are shown in Figures 35, 36 and 37. The manufacturing steps for the toothbrush handle are shown in Figures 38 and 39.

Toothbrush head manufacturing

Dried and split bamboo wedges are sourced from cooperatives in Musanze as they have equipment capable of these steps and bamboo wedges take less space than raw bamboo culms when transported.

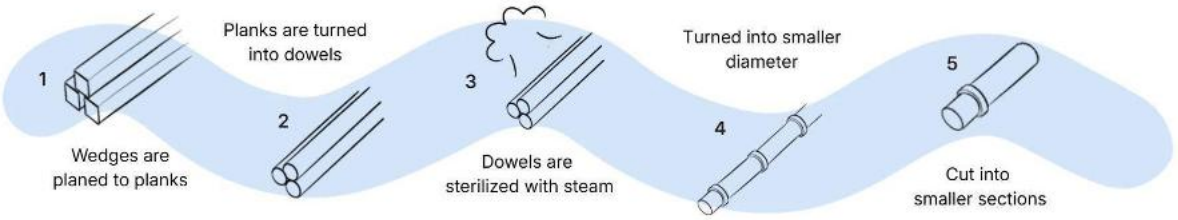


Figure 35. Manufacturing of concept 2, toothbrush head, steps 1-5

1. The wedges are planed with a thickness planer into square planks.
2. The planks are turned into dowels with a lathe.
3. The dowels are treated with pressurized hot steam.
4. The dowels are turned again, but certain sections of the dowel are left at the initial diameter.
5. The dowel is cut into smaller sections, where a larger diameter is kept close to one end.

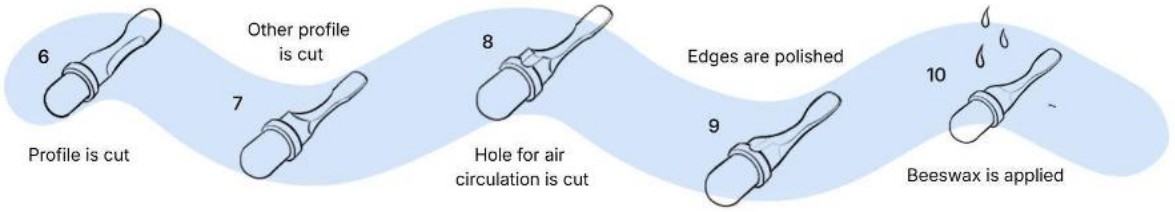


Figure 36. Manufacturing of concept 2, toothbrush head, steps 6-10

6. The smaller sections are cut with a router to create the shape of the toothbrush head in one profile.
7. The side is cut with a router to create the shape of the toothbrush head in the other profile.
8. A cut is made with a router to create the hole between the toothbrush head and handle.
9. The sharp edges are polished.
10. A beeswax is applied as a protective coating. The coating is melted and applied with a cloth. The toothbrush is then left to cure for 24 hours at room temperature, to let the wax solidify.

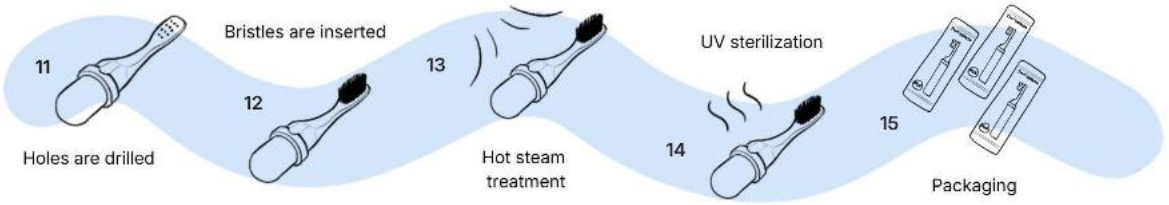


Figure 37. Manufacturing of concept 2, toothbrush head, steps 11-15

11. Holes are drilled in the head with a drilling machine.
12. Bristles are inserted in the drilled holes with a tufting machine.
13. Dust from shaping is removed from the toothbrush handles.
14. Toothbrushes are sterilized for 30 minutes in a UV-C box.
15. The toothbrush top and handle is assembled and packaged.

Toothbrush handle manufacturing

Dried bamboo culms with a smaller diameter are sourced from Musanze.

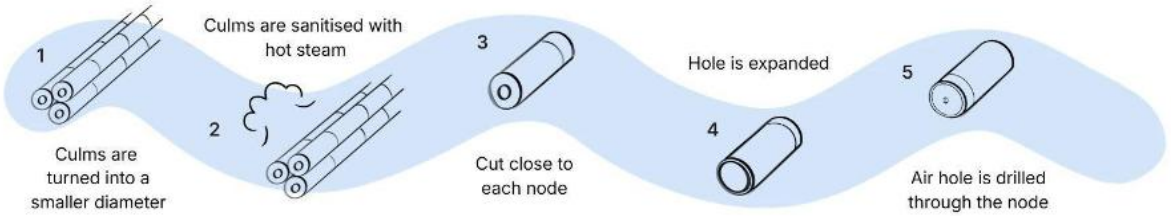


Figure 38. Manufacturing of concept 2, handle, steps 1-5

1. Culms are turned with a lathe to the diameter of the handle.
2. The turned culms are treated with pressurized hot steam.
3. The bamboo culms are cut to smaller sections, with one edge close to a node.
4. The natural bamboo holes in the cross section are expanded with a pillar drill, down to, but excluding the node close to the edge. This is done on both sides.
5. A small hole is drilled though the internode.

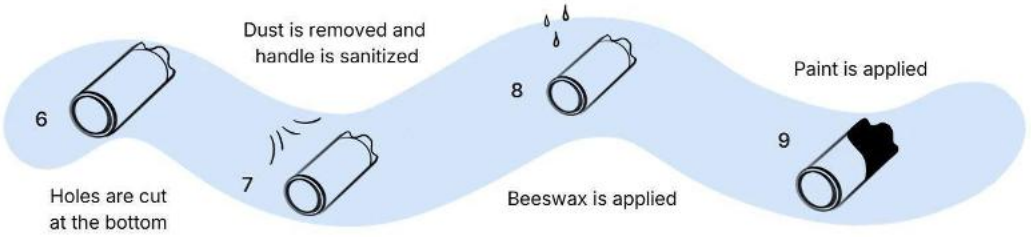


Figure 39. Manufacturing of concept 2, handle, steps 6-9

6. Holes are cut with a router at the bottom of the handle.
7. Dust from shaping is removed and the toothbrush handles are then sanitized using hot water or steam. The products are then dried in a controlled environment.

8. A beeswax is applied as a protective coating. The coating is melted and applied with a cloth. The toothbrush is then left to cure for 24 hours at room temperature, to let the wax solidify.
9. Paint is applied on the toothbrush handle, by dipping it in paint.

7.3 Packaging

Both toothbrush models are packaged in un-bleached kraft paper pouches. The front of the package displays the appropriate age for each toothbrush model and indicates that the toothbrush is made in Rwanda. On the back of the pouch, the user is informed about how to dispose of both the toothbrush and packaging, and how often to replace the toothbrush. The packages of the concept 1 toothbrushes also include a note informing the user to use a non-toxic permanent marker when writing the child's name. Kraft paper was chosen as the packaging material because of its biodegradability, and the pouch format was selected to save space in transports. Figure 40 shows mockups of the toothbrush packaging.



Figure 40. Packaging mockups (print design created in procreate, image of paper pouches generated with Leonardo.Ai and refined with photoshop, prompts used for the image generation can be found in Appendix I).

7.4 Prototypes

To further visualize the toothbrush concepts and communicate the design to project members and stakeholders, physical prototypes were created. Concept 1 was prototyped using Arundinaria alpina bamboo, worked in the wood workshop at KTH campus. Pieces of bamboo were sawn and polished to the decided shape using a sanding belt. The handle was then treated with beeswax from Rwanda, and "Izina" was engraved using a laser engraver. Modelling foam was shaped, painted and attached to the handles to visualize the bristles. The finished prototypes of concept 1 are shown in Figure 41.



Figure 41. Prototypes of Concept 1.

Concept 2 was prototyped using 3D printing and spray paint. Bristles were inserted by hand in one of the models to better communicate the final appearance. The finished prototypes of concept 2 are shown in Figure 42.



Figure 42. Prototypes of concept 2.

7.5 Implementation

The oral health intervention is meant for implementation in a school based oral health program, playing a part in a system of several stakeholders, creating value in multiple stages of the process. Figure 43 shows a simplified illustration of the proposed product system, and the different steps are explained below.

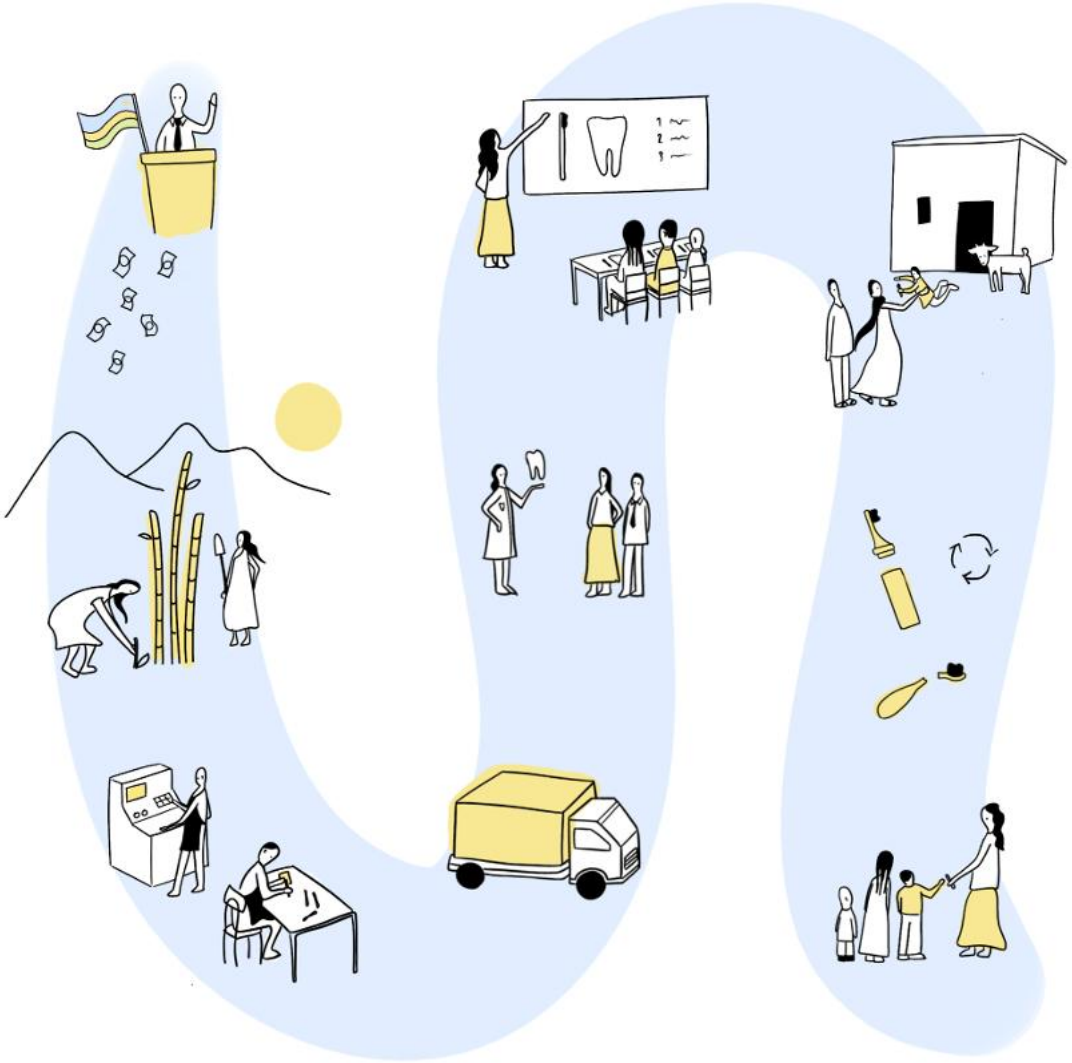


Figure 43. Implementation roadmap

Government

In the first image in the top left corner of the roadmap, the government can be seen investing in the project. One of the key outcomes of the overarching project's advocacy work is a government commitment to implement a school-based oral health program in primary schools. By investing in the locally manufactured toothbrushes for distribution within the program, the government adheres to the national oral health strategies, decreases dependency on imported toothbrushes and supports local entrepreneurship, specifically in the local bamboo and apiculture industries. While the government will be the biggest customer and provide a steady stream of funds after implementation, upfront investments in machines and salaries will be covered by external investors.

Material suppliers
At the next step in the roadmap, materials are being supplied. Farming cooperatives growing bamboo in the buffer zones around volcanoes national park provide Arundinaria Alpina bamboo culms, and beekeeper cooperatives provide beeswax. The nylon 11 bristles are imported from a chinese supplier, since there are no local castor oil nylon manufacturers.

Manufacturing
The toothbrushes are then manufactured in collaboration with local workshops. Already existing tools for drying, cutting and splitting the bamboo are used for the first steps of the manufacturing process. The remaining machines are provided by the project. The workshop staff is trained to operate the new machines. ARECO Rwanda Nziza, a government-funded organisation with roots in both the agricultural and manufacturing communities, is included as a project partner organising contacts with both bamboo suppliers and local workshops.

Transport
From the workshops, toothbrushes make their way to schools across the country. Logistics are managed to ensure every participating school receives a reliable supply.

Dentists and teacher training
Before the oral health program is launched in the schools, teachers go through training with dentists. The sessions follow a clear curriculum developed by dentists for the program, so that teachers are prepared to educate the students.

Schools
As part of the oral health program, each child receives a toothbrush, takes part in fluoride rinsing, and gets taught the basics of oral health. Every child writes their names on the brush to avoid mix-ups, and the brushes are kept at school between uses. After implementation of the program, brushing teeth after lunch is part of the daily school routine.

Behavior spillover
The new habits and knowledge of the children will affect the families creating a positive behavior spillover.

Toothbrush replacement
Every third month, the toothbrush head is broken off and the toothbrush is disposed of, or the toothbrush head is disposed of, depending on which model is implemented. The children will then get a new toothbrush, or toothbrush head, in school.

8. Discussion & next steps

This chapter presents our reflections on the project as a whole: what could have been improved, ethical considerations, and what comes next.

8.1 Discussion

In this section, the developed proposal is evaluated in relation to the aims and objectives set at the beginning of the project. Material choices and manufacturing trade-offs, reflection on the methods, the implementation roadmap, and local ownership are also discussed.

8.1.1 Material trade-offs

The objectives in regard to material selection were largely fulfilled, as bamboo and beeswax were identified as locally available and suitable materials for the toothbrush handle and coating. However, some material decisions required balancing local availability with the need for a reliable and durable toothbrush, especially in the choice of bristle material. This reflects the material trade-offs identified in the literature study, where natural materials were often more sustainable but less effective and predictable in terms of hygiene, durability and cleaning performance.

Using nylon 11 as the bristle material goes against the project objective of local material availability. Natural materials were therefore thoroughly researched before selecting a non-biodegradable, imported bristle material. According to the research reviewed, most natural options, such as plant fibre or boar hair, could not match the cleaning performance of nylon. Chewing sticks and miswaks were the only exceptions found, as they performed comparably to nylon, but required proper preparation and technique to be effective, which in itself presented a risk of misuse. Since it was of high importance that the toothbrush should be easy to use, these alternatives were rejected. Compromising the function of the brush was therefore not considered acceptable within the context of a school-based intervention. Nylon has dominated the market for decades, and both the timeframe of the project and the limits of our expertise as design engineering students made it unrealistic to attempt developing and evaluating an entirely new bristle material. However, Rwanda's manufacturing sector is growing, and developments are also taking place in the biodegradable toothbrush sector. Locally produced and biodegradable bristles may therefore become available in the future. Until then, nylon 11 was selected as the best compromise at this stage, as it is partly bio-based and was considered more reliable than the natural alternatives. The simple bristle arrangement chosen for both concepts aligns with the literature reviewed in Section 2.5.1, where several studies indicated that more complex bristle configurations do not necessarily provide enough additional benefit to justify higher costs in school-based and cost-sensitive contexts.

The coating choice involved a different type of trade-off. Unlike the bristle material, beeswax fits the project aim well because it has good local availability and aligns with the sustainability ambitions of the project. The uncertainty is instead related to technical performance, since durability testing of prototypes treated with beeswax will be necessary to evaluate whether it provides enough protection during repeated use. If needed, future work could investigate mixing the wax with oils to help the coating penetrate the material deeper and improve durability. Moringa oil is produced locally in Rwanda and has good oxidation stability, while tung oil is not locally produced but has strong hardening and moisture resistance qualities. The trade-off is that a blended coating formula would add complexity, both in sourcing and production. Therefore, a

pure beeswax coating should be used if prototype testing shows sufficient durability, since it is the most practical and appropriate choice within the scope of the project.

8.1.2 Manufacturing feasibility and cost

The manufacturing objective was addressed through the development of a locally adapted production setup, but its feasibility at a larger scale remains uncertain. Since automated large-scale machines were substituted by smaller machines requiring more manual labour, the expected production capacity is lower than in conventional toothbrush manufacturing. This setup is therefore not intended as a final solution for national-scale production, but rather as a starting point to evaluate local production capabilities and as a proof of concept for potential investors, including the Rwandan government. If higher production capacity for concept 1 is needed, the process could also be scaled by adding more units at bottleneck steps or by gradually transitioning toward a more automated setup, while limiting financial loss from the initial smaller-scale investment.

The lower level of automation also creates an economic trade-off. The smaller-scale machines reduce the initial investment cost and make the setup more realistic for early implementation, but the actual cost per toothbrush and break-even point remain unknown. A full cost estimation was not possible within the scope of this project, since several variables remained uncertain, including electricity costs, wages, maintenance, material waste and transportation. Making calculations based on uncertain assumptions risked giving a misleading impression of the economic feasibility of the proposal.

Further feasibility studies are therefore needed before implementation. These should be conducted together with ARECO Rwanda Nziza and potential local production partners in order to estimate real production costs, evaluate bottlenecks and determine how the process could be scaled. If concept 1 is later transitioned into concept 2, some of the most important machinery investments, such as drilling and tufting equipment, could still be used, which reduces the financial risk of starting with a smaller-scale setup.

8.1.3 Reflection on methods and data quality

Methods used during this project provided insights into both the local context and the technical requirements for the proposed toothbrush. Interviews and site visits were especially important for understanding locally available materials, possible production partners and the broader stakeholder system for the project. The findings are however based on a limited number of visits and stakeholder interactions and they should therefore not be seen as full representations of all possible production conditions or implementation contexts.

The questionnaires resulted in useful information about schoolchildren's oral hygiene habits and preferences, but the data also has some limitations. Questionnaires were chosen because they made it possible to collect quantitative data from a large number of children, complementing the more qualitative data gathered through interviews with other possible stakeholders. The questionnaire consisted mainly of multiple-choice questions, with some questions also allowing

written answers. Interviews with children could likely have yielded more detailed responses to open-ended questions, as respondents would not have been limited by the available writing space and follow-up questions could have been asked. Interviews would however not have been possible to conduct at the same scale as the questionnaires. Additionally, interviews would have been difficult to conduct as there would be a language barrier. The language barrier was possible to circumvent through questionnaires, since we could get them properly translated. Interviews would have required us to have a translator assisting us during all interviews, and there would still be a risk of miscommunication. For interviews to have been conducted properly, more time than available would be necessary.

The stakeholder map and stakeholder value network helped structure the actors, roles and value exchanges within the proposed system. However, these methods are interpretative and based on the information available during the project. Further validation with local stakeholders would therefore be needed before using them as a basis for implementation.

8.1.4 Feasibility and challenges of implementation roadmap

The implementation roadmap provides a possible structure of how the proposed toothbrush could be introduced in a school-based oral health program. The roadmap connects material sourcing, manufacturing, distribution, teacher training, use in schools and replacement of toothbrushes and helps translate the product proposal into a broader implementation system. Since the toothbrush is not intended to function as an isolated product, a structure for the larger oral health intervention is necessary.

The roadmap should however still be understood as a conceptual proposal rather than a validated detailed implementation plan. Multiple practical challenges remain, including funding, logistics and teacher involvement. These factors are especially important if the intervention is scaled to a nationwide program, since continuous production and distribution would be required for the system to remain functional over time.

A central challenge for this intervention is therefore long-term responsibility. For the roadmap to be feasible, it must be clear which exact actors are responsible for financing, production, distribution, training and follow-up. Without clear ownership of these steps, the intervention risks becoming difficult to maintain. Clear ownership is therefore necessary for the roadmap to become practically feasible and sustainable over time.

The proposed implementation roadmap also reflects findings from the literature study on school-based programmes. Both the Swedish school based oral health programme and Rwanda's National School Feeding Programme show that schools can work well as platforms for preventive health interventions when routines, distribution and institutional responsibility are clearly organised. This supports the decision to treat the toothbrush not only as a product, but as part of a broader system including production, distribution, school use and replacement.

8.1.5 External perspective and local ownership

Something worth mentioning is the inherent complexity of us being two Swedish students designing a Rwanda-branded product, specifically for the Rwandan cultural and economic context. As Swedish students, our perspectives are shaped by experiences and assumptions which differ significantly from those of the users this project is meant for. During the field studies, we tackled this by trying to involve as many local partners and users as possible, while trying to avoid making assumptions and stay open-minded.

A common critique of Design for the Base of the Pyramid is that the framework itself risks fostering a paternalistic relationship and, in the end, decreasing the independence of local communities [95]. It is important that Rwanda owns both the production and the product itself for this project to be sustainable in the long run. That would require building local competence to manage production and future development of the product, and ensuring that decisions are made by people and organisations closer to the problem instead of creating a dependency on external parties, like the project team. Involving ARECO Rwanda Nziza and including training of local entrepreneurs is a good start to make sure both the product and the product system can run and evolve without the external project members in the future.

Overall, the proposal should be understood as a grounded concept and system proposal rather than a finished implementation plan. The project provides a starting point for further development. There are however technical, economic and organisational aspects that need to be validated, together with local stakeholders before implementation.

8.2 Conclusion

In this project, two proposed toothbrush concepts largely made of locally available materials, designed for implementation in a school based oral health program, and a proposed implementation roadmap have been generated.

The bamboo species Arundinaria alpina was identified as the most suitable material for the toothbrush. Arundinaria alpina has good local availability, is biodegradable, and aligns with the government's current ambition to expand the local bamboo industry. The fact that arundinaria alpina is commonly grown in buffer zones close to national parks, and close to rivers to prevent erosion also adds the possibility for the project to create additional value by supporting local ecosystems and entrepreneurship. Beeswax, sourced from local beekeeper cooperatives, was chosen as the protective coating for the toothbrush and like bamboo, adds possibility for local value creation through supporting local entrepreneurship. The material selected for the bristles was nylon 11, which is not produced locally but has a lower CO2 impact than conventional nylon and is a well-tested and effective bristle material, ensuring that the implemented toothbrush will fulfill its main function: to clean teeth.

Two proposed toothbrush designs were created based on user studies, site visits, interviews and desktop research. The proposals enable gradual scalability in the project, as one design is meant for early stages of the project, requiring minimal resources and utilising already existing tools and

skills, and the second design is created for implementation in a later stage of the project, when funds are secured.

Two proposed manufacturing setups and plans were created for the two concepts. One setup is planned within a partnership with local workshops, keeping initial investments manageable. The second manufacturing proposal involves setting up a new factory in Kigali special economic zone- suited for large scale production and high production capability. Finally, a roadmap for the overall intervention was created, covering sourcing of the materials, manufacturing and distribution of toothbrushes in schools. This roadmap provides a suggested, but brief, outline for the project and all its involved steps.

In conclusion, the project aims of identifying locally available and sustainable materials, developing a proposal for a locally adapted manufacturing process, designing a user-friendly toothbrush for schoolchildren in Rwanda, and outlining the product system surrounding the toothbrush were largely met, with some compromises along the way, mainly the bristles not being locally sourced and biodegradable. Overall, this project demonstrates that producing toothbrushes locally in Rwanda for implementation in school based oral health programs is not only possible, but feasible. Throughout our field studies, we were met with enthusiasm and a will to cooperate from local actors. There is also a strong interest from the government, as the project aligns with multiple national strategies. The project team of the overarching project will find both support and motivation as they continue the work on the implementation of the school based oral health program and toothbrush manufacturing.

8.3 Continued work

This project consists of the foundation of setting up a production for a toothbrush to distribute within Rwanda's national oral health program. However, there is still a long way to go until the toothbrushes actually reach the hands of the Rwandan school children. This section includes comments and recommendations on how the project members can take the project further when our work is done.

- Policy advocacy and government engagement

The advocacy work carried out by the project members to secure government commitment to the school based oral health program is the key enabler of this project. Continued engagement with the Rwandan government should be in focus throughout the following stages.

- Feasibility study

Using the contact with ARECO provided through these field studies, a feasibility study should be carried out investigating availability of the local bamboo, prototyping with the chosen materials and mapping of suppliers, manufacturing partners and potential investors. The study should also put focus on estimating involved costs to build a cost model based on real data.

- Development and testing of proposal

Once the feasibility study is done, the prototypes should be tested and evaluated. Durability tests of the coating should be carried out as well as small scale tests in schools to assess functionality before initiating production. REMA should be contacted to assess the need for clinical testing of the finished toothbrush, to strengthen the credibility and increase the likelihood of government adaptation.

- Local ownership

After setting up production, continued work should focus on gradually transferring knowledge and responsibility to local actors. This is done by training the staff and involving Rwandan engineers in future development of the toothbrush. The goal should be a product and product system fully owned by Rwanda.

- Scaling

Finally, as implementation of the program is getting closer, work will be needed to scale the production to meet the demands for national distribution, navigating public procurement and making sure all requirements are met. Close collaboration with the Rwandan government and Ministry of Health will be essential at this stage.

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APPENDIX A - Interview guides

ARECO

[intro: inform about project and ask for permission to record interview]

Tell us more about your organisation and what you do!

How do you work specifically with the bamboo and forestry industry?

Can you tell us more about the bamboo industry in Rwanda?
Follow-ups:
How much bamboo is grown,
Which kinds are grown,
What are the main uses of bamboo in Rwanda,
Where is bamboo being grown commercially?
Are there facilities for preparing the bamboo (cutting or pressing) or is the bamboo normally sold in "whole logs"?

Are there any downsides of the bamboo industry?
(biodiversity, price, working conditions etc)

Which kinds of locally grown wood are available in Rwanda?
(We have read that 80% of the forest industry consists of eucalyptus wood, but that it's bad for biodiversity- do you know anything about this?)

Which material do you think would be the most environment friendly and easy to get hold of for a large scale production of toothbrush handles?

Regarding "Project to Support the Promotion of Bamboo for Environmental Protection, Poverty Reduction, and Climate Change Mitigation in the Volcanoes National Park and Rugezi Swamp Areas"

- How does it add to the poverty reduction in the rural areas surrounding Volcanoes National Park?
- Was the bamboo grown in this project harvested or was it mainly for climate change mitigation?

Regarding "ARECO Participates in Nyabihu and Musanze Open Days to Highlight Community Engagement and Environmental Initiatives"

- "One of the highlights was a display of handcraft items made from bamboo, produced by local women's cooperatives that have been supported by ARECO and its partners."
  Where was the bamboo produced? And was it produced as part of this project or was the production of bamboo already taking place? And do you know what type of bamboo it was?

SOS Childrensvillages

[intro: inform about project and ask for permission to record interview]

How have you worked with SOS Children's villages? (What's your role/roles)

What is your perception of the current oral health situation in Rwanda - we have read the national oral health survey from 2017 - how has the situation changed since then?

Which methods, except from normal toothbrushing, are the most common ways of cleaning the teeth here?

Do you know what have been the most common methods historically?

What would you say are the gaps/hurdles hindering the people that's not brushing their teeth from doing so? Is it mainly financial reasons or are there other reasons?

Were you involved in the school based oral health project conducted by SOS and dentists without borders 2012-2018? If so- what were your learnings from this project?

Would you say that people are hesitant towards using fluoride?

In your experience, what materials are children's toothbrushes made from in Rwanda?

How would you describe the accessibility of toothbrushes in rural areas?
(Do you know if the price of toothbrushes is a big hurdle)

Do you have any insight on what the view is in regards to materials in toothbrushes? For example, are plastic toothbrushes more popular than bamboo toothbrushes, or vice versa?

As a dentist- are there any aspects of the toothbrush design that you think is extra important?

Arth Biobag

[intro: inform about project and ask for permission to record interview]

What products do you manufacture with your biodegradable plastic?

Would the material still be compostable with thicker products (like a toothbrush handle)?

How does biodegradability affect durability? Is there a risk that a product starts degrading before the intended EOL?

What is the material made from?

From where do you source the corn starch, sugar cane etc?

Tell us about the manufacturing process, from raw material to finished product! How does it differ from conventional plastic manufacturing?

Do you export the products/material to other countries? Is it mainly used in Kigali or other cities as well?

How is your business affected by the single use plastic ban in Rwanda? (Tell us more about the plastic ban!)

What are the prices on your plastic compared to conventional plastic?

Do you only sell finished products, or do you also sell the base material (granules)?

Do you think biodegradable plastic would be a feasible material for a sustainable toothbrush handle?

Ultimate Forests Ltd

[intro: inform about project and ask for permission to record interview]

Tell us more about your role at Ultimate Forests!

Tell us more about how Ultimate forests operate.

Which tree species do you primarily cultivate, and why?

How do you integrate sustainability principles into your operations (e.g., replanting, biodiversity, soil protection)?

Do you produce eucalyptus wood?
If so: As you are a sustainable company, how does the eucalyptus wood industry affect biodiversity?

Which types of companies/customers do you normally sell wood to?

Which type of wood would you recommend for making a sustainable and affordable toothbrush?

- Why?
- Tell us about the mechanical properties, moisture absorption etc.
- What is the annual production of that wood?

Do you have experience processing your wood for smaller consumer products (e.g., furniture components, handcrafted items, prototypes)?

What kind of equipment or expertise is required to transform your wood into toothbrush handles?

Can you suggest suitable surface treatments or finishes that are safe for consumer use (e.g., water-based oils, natural waxes)?

Do you have insights on how local communities perceive wood-based consumer products compared to plastic ones?

Rwanda Wood Value Chain Association

[intro: inform about project and ask for permission to record interview]

Tell us more about the Wood Value Chain Association!

Which type of wood would you recommend for making a sustainable and affordable toothbrush?

- Why?

How much does that wood cost?

Kontakta dem om vi vill genomföra, de kan fixa good price och se till att mer bambu planteras

Do you know of any products similar to this toothbrush handle, that are made out of wood?

How is it made?
By who?

Do you know if there are companies using CNC routers for wood products in Rwanda currently? Who?

What kind of equipment or expertise do you think is required to transform your wood into toothbrush handles?

Can you suggest suitable surface treatments or finishes that are safe for consumer use (e.g., water-based oils, natural waxes)?

What do you know about the bamboo industry in Rwanda, and in what ways is it different from the wood industry? (Is there, in your opinion, enough bamboo to provide for a large scale production of toothbrushes)

RDB

[intro: inform about project and ask for permission to record interview]

Which material aligns best with the government's current strategies and policies?

Are there tax exemptions or reliefs on any of the two materials?

Do they have any valuable contacts within bamboo products manufacturing?

Which of the following do you think is best:

- Starting a new factory, to which machines for the toothbrush manufacturing are brought?
- Hiring the production from a current company, if there is one that has some existing ?
- machines and/or relevant knowledge for toothbrush manufacturing?

Relief and development support organization

[intro: inform about project and ask for permission to record interview]

Village:
How many live in this village?
What is the main source of income?
What are the main challenges for the families here?
How is the access to healthcare?
And access to clean water?
What can you say about the dental health in the village?
Describe daily life!

The school:
How many children attend this school?
How many teachers?
How big are the classes?
Describe a normal day for school children here!
Do children learn about toothbrushing in school?

Toothbrushing:
How many of the kids would you say own their own toothbrush?
How often do the kids normally brush their teeth?
Do they brush in school or at home?
Where do the families usually buy their toothbrushes?
Where are toothbrushes sold in smaller villages?
What is the most common kind of toothbrush here?

What do families do if they can't afford toothbrushes? (chewing sticks, sharing toothbrushes?)
Do you have any idea about what could motivate the kids to brush their teeth?
Do you think a wooden/bamboo toothbrush or a plastic one would be more popular? Why?

"If you could change one thing to improve children's oral health in this community, what would it be?"

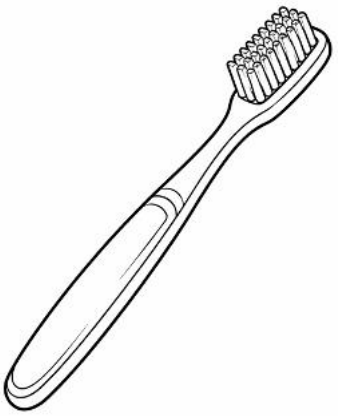
APPENDIX B - Questionnaires in English

Questionnaire for school children aged 9-16 in Rwanda.

A. Demographic characteristics of school children

1. How old are you?

a) 9
b) 10
c) 11
d) 12



2. Are you a girl or a boy?

a) Girl
b) Boy
c) Prefer not to say

3. Which class are you in?

a) P4
b) P5
c) P6

4. What's the name of your school?

Please write: _____

B. Information on school children's brushing habits and preferences

5. Do you own a toothbrush?

a) Yes
b) No
c) Don't know

6. If yes, what kind of toothbrush do you own?

a) Plastic toothbrush for kids
b) Plastic toothbrush for adults
c) Wooden/Bamboo toothbrush for kids

d) Wooden/Bamboo toothbrush for adults
e) Chewing stick
f) Other, please specify: _____
g) I don't know

7. If yes, how often do you replace your toothbrush?

a) Every 0-3 months
b) Every 3-6 months
c) Every 6-12 months
d) Less often

8. Have you ever seen a wooden toothbrush?

a) Yes
b) No
c) I don’t know

9. If you got to choose, which toothbrush would you prefer?

a) A plastic toothbrush
b) A wooden toothbrush
c) I don’t care
d) I’m not sure

10. How often do you brush your teeth?

a) Two times each day
b) Once a day
c) Less than once a day
d) Never

11. When do you usually brush your teeth? (you may select more than one alternative)

a) Before breakfast
b) After breakfast
c) After lunch at school
d) After dinner
e) Before bedtime
f) Other time, please specify: _____

12. Where do you store your toothbrush in between uses?

a) In a cup/mug (standing up)
b) Laying down on the sink or on shelf

c) In a toothbrush case
d) Other, please specify: _____
e) I don’t know

13. Which of these statements do you agree with? (you may select more than one alternative)

a) I like brushing my teeth
b) I think brushing my teeth is boring
c) I think brushing my teeth is important for my overall health
d) I think brushing my teeth does not affect my overall health

14. Do you ever feel pain in your gums when you brush your teeth?

a) Yes, every time
b) Yes, sometimes
c) No, never
d) I don't brush my teeth

15. Do you think it’s easy to reach all your teeth with your toothbrush?

a) Yes
b) Sometimes
c) No
d) I don’t have a toothbrush

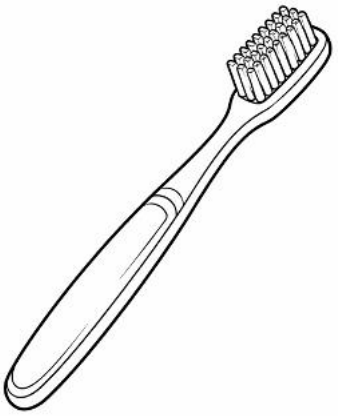
16. What would make toothbrushing more fun?

a) A colorful toothbrush
b) A toothbrush with my name on it
c) Other: _____
d) I don’t know

APPENDIX C - Questionnaires in Kinyarwanda

Urutonde rw’ibibazo bigenewe abana b’abanyeshuri bafite imyaka 9-16 mu Rwanda.

A. AMAKURU RUSANGE KU MWANA



1. Ufite imyaka ingahe?

a) 9

b)10

c) 11

d)12

2. Uri umukobwa cyangwa umuhungu?

a) Umukobwa

b) Umuhungu

c) Nahitamo kutabivuga

3. Wiga muwuhe mwaka

a) Mu wa kane

b) Mu wa gatanu

c) Mu wa gatandatu

4. Tubwire izina ry´ikigo wigaho

Ryandike hano_____

B. AMAKURU KU BIREBANA NO KOZA AMENYO KWAWE N'IBYO UKUNDA.

5. Waba ufite uburoso bwo koza amenyo?

a) Yego

b) Oya

c) Ntabyo nzi

6. Niba ufite uburoso bwo koza amenyo, ni ubuhe bwoko ufite?

a) Uburoso bwa plasitike bwagenewe abana

b) Uburoso bwa plasitike bwagenewe abantu bakuru

c) Uburoso bukozwe mu giti cyangwa mu mugano bwagenewe abana

d) Uburoso bukozwe mu giti cyangwa mu mugano bwagenewe abantu bakuru

e) Agati

f) Ibindi _____ (bivuge)

g) Ntabwo mbizi

7. Niba ufite uburoso bwo koza amenyo, ubuhindura nyuma y'igihe kingana iki?

a) Nyuma y'amezi 0-3

b) Nyuma y'amezi 3-6

c) Nyuma y'amezi 6-12

d) Nyuma y'umwaka

8. Wigeze ubona uburoso bwo koza amenyo bukozwe mu giti?

a) Yego

b) Oya

c) Ntabwo mbizi

9. Ni ubuhe bwoko bw'uburoso bwo koza amenyo wahitamo?

a) Uburoso bwa plasitike

b) Uburoso bukozwe mu giti/mu mugano

c) Ubwo aribwo bwose

d) Ntabwo mbizi

10. Woza amenyo kangahe?

a) Kabiri buri munsi

b) Rimwe ku munsi

c) Munsi ya rimwe ku munsi (rimwe na rimwe)

d) Sinjya noza amenyo

11. Ni ryari woza amenyo ubusanzwe? (ushobora guhitamo ibisubizo birenze kimwe)

a) Mbere yo gufata ifunguro rya mu gitondo

b) Nyuma yo gufata ifunguro rya mu gitondo

c) Nyuma y´ ifunguro rya saa sita ndi ku ishuri

d) Nyuma yo gufata ifunguro rya ni mugoroba

e) Mbere yo kuryama

f) Ikindi gihe: _____ (kivuge)

12. Iyo utari gukoresha uburoso bwawe bwo koza amenyo ububika he?

a) Mu gikombe cyangwa mu kirahuri buhagaze

b) Mburambika kuri lavabo cyangwa mu kabati/kuri etagere

c) Mu kantu babuguzemo

d) Ahandi: _____(havuge)

e) Ntabwo mbizi

13. Mu nteruro zikurikira, hitamo izo wemera (ushobora guhitamo izo wemera zose)

a) Koza amenyo ndabikunda

b) Koza amenyo mbona birambirana

c) Ntekereza ko koza amenyo yanjye ari ngombwa kugirango ngire ubuzima bwiza

d) Ntekereza ko kutoza amenyo ntacyo byakwangiza ku buzima bwanjye muri rusange

14. Ese ujya wumva ubububare mu ishinya igihe woza amenyo?

a) Yego, buri gihe ndababara

b) Yego, rimwe na rimwe ndababara

c) Oya, nta na rimwe mbabara

d) Sinjya noza amenyo

15. Ese utekereza ko byoroshye kugeza uburoso ku menyo yawe yose mu gihe urimo koza amenyo?

a) Yego

b) Rimwe na rimwe

c) Oya

d) Nta buroso bwo koza amenyo ngira

16. Ni iki cyatuma koza amenyo birushaho kukunezeza, ukabikora wishimye?

a) Uburoso bufite amabara meza

b) Uburoso bwanditseho izina ryanjye

d) Ikindi: _____(kivuge)

e) Ntabwo mbizi

APPENDIX D - Assent forms in English and Kinyarwanda:

English:

Hello!

We are students from KTH Royal Institute of Technology in Sweden. We are doing a school project about designing a toothbrush for schoolchildren in Rwanda.

We would like you to answer a short questionnaire about how you brush your teeth and what you like about toothbrushes.

- Taking part is your choice.
- You can skip questions or stop at any time.
- Your answers will be anonymous, which means no one will know which answers are yours.
- Your answers will only be used for our school project.

If you want to take part, please sign below.

I understand what the questionnaire is about and I agree to take part.

Name: _____

Signature: _____

Date: _____

Kinyarwanda:

ICYEMEZO CY'UMWANA CYO KWITABIRA UBUSHAKASHATSI

Muraho!

Turi abanyeshuri baturutse muri KTH Royal Institute of Technology yo muri Suède. Mu rwego rw'amasomo, turimo gukora umushinga w'ubushakashatsi ugamije kwerekana uburyo hakorwa akaroso ko koza amenyo kagenewe abana b'abanyeshuri mu Rwanda.

Turagusaba kudufasha ugasubiza ibibazo bigufi bijyanye n’uko usanzwe woza amenyo n’ibyo ukunda ku birebana n’akaroso ko koza amenyo.

Iki kiganiro kizamara iminota igera kuri 15.

Iby’ingenzi ugomba kumenya:

• Kwemera kwitabira ni ubushake bwawe. Ushobora no guhitamo kutabyemera.

• Ushobora kudasubiza ikibazo utifuza cyangwa ugahagarika igihe icyo ari cyo cyose.

• Ibisubizo byawe bizagirwa ibanga. Nta muntu uzamenya ibisubizo byawe.

• Ibisubizo byawe bizakoreshwa gusa muri uyu mushinga w’ubushakashatsi.

Niba wemera kwitabira, andika izina ryawe ushyire n’umukono hasi aha ahabugenewe.

Nsobanukiwe neza ibijyanye n’ubu bushakashatsi kandi nemeye kugiramo uruhare, nsubiza utuazo mbazwa.

Izina ry’umwana: _____

Umukono: _____

Itariki: _____

APPENDIX E - Consent forms in English and Kinyarwanda

English:

Title: Design of a locally manufactured and user-friendly toothbrush for children in Rwanda

Informed consent to parents

Greetings,

We are Ebba Zingmark and David Brimberg Tengzelius, master's students from KTH, Royal University of Technology in Integrated Product Development. Currently supported by University of Rwanda, we are conducting a study here in Rwanda on designing a toothbrush for children. The aim is to develop a proposal on the local production of user-friendly toothbrushes that can be easily distributed in primary schools for prevention of oral diseases in Rwanda.

We are inviting you to allow your child to take part in this study. We only aim to interview your child about brushing habits and toothbrush preferences.

There are no risks or direct benefits to the participants, however the study will have an impact on the potential of local production of user-friendly and cheaper toothbrushes for children in Rwanda, aiming to prevent common oral diseases among children in Rwanda. At the end of the survey, each participating child will be given a toothbrush as a thanks for the participation.

To ensure confidentiality, the names of parents/guardians and child will not be indicated on the questionnaire and during the record of data. We will replace names with codes. We are requesting your names only to know the children who are allowed to take part in the study.

The interview will take approximately 15 minutes for each child. If at any time during the interview the child wants to stop, they have the right to do so without any negative consequences.

By signing this consent form, you confirm that you understand the purpose of the study and consent that your child take part in this study.

I have read this consent form, or someone has read it for me, and I understand that there is no anticipated risk associated with this study. I voluntarily allow my child to participate in this study

Name of the parent/guardian: _____

Signature: _____

Name of the child: _____

Date: _____

Ebba Zingmark: ebbazi@kth.se
David Tengzelius: davidbt@kth.se

If you have any issue, please contact University of Rwanda Research ethics committee representative Prof Stefan Jensen: sjansenur@gmail.com

Kinyarwanda:

UMUTWE W'UBUSHAKASHATSI: Gukora igishushanyo mbonera cy'akaroso ko koza amenyo y'abana, kakorerwa mu Rwanda kandi kakorohera abana kugakoresha

URUPAPURO RW'ICYEMEZO CY'ABABYEYI CYEMERERA UMWANA KWITABIRA UBUSHAKASHATSI

Muraho neza,

Turi Ebba Zingmark na David Brimberg Tengzelius, abanyeshuri biga icyiciro cya kabiri cya kaminuza (Master's) muri KTH Royal University of Technology mu gihugu cya Suwede, mu ishami rya *Integrated Product Development*. Tubifashijwemo na Kaminuza y'u Rwanda, turi gukora ubushakashatsi hano mu Rwanda bugamije gukora igishushanyo mbonera cy'akaroso ko koza amenyo kagenewe abana.

Intego y'ubu bushakashatsi ni ukwerekana uburyo hakorwa akaroso ko koza amenyo kagakorerwa mu Rwanda, kakorohera abana kugakoresha, kandi kakabasha gukwirakwizwa mu buryo byoroshye mu mashuri abanza, mu rwego rwo gukumira indwara zo mu kanwa n'amenyo ku bana bo mu Rwanda.

Twagusabaga rero ko wakwemerera umwana wawe kwitabira ubu bushakashatsi. Umwana tuzamubaza ibibazo byoroheje bijyanye n'uko asanzwe yoza amenyo n'ubwoko bw'akaroso ko koza amenyo akunda.

Ingaruka n'inyungu zishoboka

Nta ngaruka mbi zizaterwa no kwitabira ubu bushakashatsi ziteganyijwe.

Nubwo nta nyungu zihuse ku mwana ku giti cye ziteganyijwe, ubu bushakashatsi bushobora gutanga umusanzu mu guteza imbere ikorwa ry'akaroso ko koza amenyo gakorerwa mu Rwanda, korohera abana kugakoresha, kandi kagura make. Ibyo byose bishobora gufasha mu gukumira indwara zo mu kanwa n'amenyo ku bana.

Nk'ishimwe ryo kwitabira ubu bushakashatsi, buri mwana uzitabira azahabwa akaroso ko koza amenyo nyuma yo kuzuza ibibazo.

Ibanga no kubungabunga umutekano w'amakuru

Amakuru azakusanywa muri ubu bushakashatsi azabikwa mu ibanga rikomeye.

- Amazina y'umwana cyangwa ay'umubyeyi/umureri ntazandikwa ku rupapuro rw'ibibazo.
- Ahubwo hazakoreshwa kode zisimbura amazina.
- Hano impamvu dusaba amazina ni ukugira ngo tumenye ababyeyi bemereye abana babo kwitabira ubushakashatsi. Gusa, uru rupapuro rubikwa ukwarwo ntabwo ruba rwegeranye n'iziriho amakuru.

Amakuru azakusanywa azakoreshwa gusa mu bushakashatsi, kandi ntazashyirwa ahagaragara mu buryo bushobora gutuma umuntu watanze amakuru amenyekana.

Uko ubushakashatsi buzakorwa

Umwana wawe azitabira ikiganiro kigufi kizakorwa n'abashakashatsi. Turateganya ko iki kiganiro cyazamara nk'iminota 15.

Kwemera ko umwana wawe yitabira ubu bushakashatsi ni uburenganzira bwawe bwite. Ushobora kwanga cyangwa guhagarika ko umwana wawe akomeza kwitabira igihe icyo ari cyo cyose, kandi nta ngaruka mbi byabagiraho.

N’umwana ubwe ashobora guhitamo kudasubiza ibibazo bimwe cyangwa guhagarika ikiganiro igihe cyose abyifuza.

Kwemera ko umwana yitabira ubushakashatsi

Gushyira umukono kuri uru rupapuro bisobanura ko:

- Wasobanuriwe neza intego n’uburyo ubu bushakashatsi buzakorwa
- Wahawe umwanya wo kubaza ibibazo no gusobanurirwa
- Wemeye ku bushake ko umwana wawe yitabira ubu bushakashatsi

Nasomye ibikubiye muri uru rupapuro rw’icyemezo, cyangwa hari uwabinsomeye abinsobanurira neza ku buryo nabisobanukiwe. Namenye ko nta ngaruka mbi ziteganyijwe muri ubu bushakashatsi, nicyo gitumye nemera ku bushake ko umwana wanjye yitabira ubu bushakashatsi.

Izina ry’umubyeyi/umurezi: _____

Umukono: _____

Izina ry’umwana: _____

Itariki: _____

Abashakashatsi

Ebba Zingmark
Email: ebbazi@kth.se

David Brimberg Tengzelius
Email: davidbt@kth.se

Uramutse hari ikibazo ugize, wamenyesha uhagarariye Komisiyo ishinzwe imyitwarire myiza mu bushakashatsi muri Kaminuza y’u Rwanda, Koleji y’Ubuvuzi n’Ubuzima

Prof. Stefan Jansen
Email: sjansenur@gmail.com

APPENDIX F - Dimensions of toothbrushes

Toothbrush	**Age**	**Total length [cm]**	**Head width (bristle width) [cm]**	**Head length (bristle length) [cm]**	**Bristle Length [cm]**
Gumpearl x Moomin toothbrush bamboo kids	3-12	14.5	1.2 (0.8)	2.6 (2)	1
Rutines Slim Comfort kids toothbrush Bamboo Ultra Soft	4-10	14.3	1.1 (0.8)	2.2 (1.6)	0.9-1.1
Humble Brush Kids Toothbrush Extra Soft Purple	2-8	14.4	1.2 (0.8)	2.5 (2)	0.9-1.1
TePe Kids Extra Soft	6+	17.1	0.95 (0.8)	2.2 (2)	0.9
Humble Toothbrush with 3 replaceable brushheads Soft	12+ (adult)	18.7	1.2 (0.9)	3.1 (2.7)	0.9-1.1
Humble Plant-based plastic Toothbrush Soft Proud	12+ (adult)	18.7	1.2 (0.9)	3.1 (2.7)	0.8-1.1
Colgate Double Action Medium	12+ (adult)	18.4	1.2 (0.9)	3.9 (3.5)	0.9-1.1
Unbranded toothbrush for travel	12+ (adult)	17.3	1.3 (0.9)	3.3 (3)	1.1
Toothbrush	12+ (adult)	18.8	1.5 (1.2)	3.4 (3)	1.1

with replaceable head bought in Rwanda					
TePe Colour Soft Toothbrush Green	12+ (adult)	18.7	1.15 (0.85)	2.6 (2.4)	1
TePe Choice Regular Soft toothbrush with 3 brushheads Pink	12+ (adult)	19	1.2 (0.9)	2.7 (2.4)	1
Humble Bamboo toothbrush with 3x Replaceable Heads Blue Soft	12+ (adult)	19	1.2 (0.8)	2.8 (2.3)	1.1
Jordan Ultralite toothbrush	12+ (adult)	19.3	1.2 (1.1)	3.1 (2.8)	0.9-1.2
Jordan Individual Reach toothbrush	12+ (adult)	19.5	1.2 (1)	2.9 (2.6)	0.9-1.1
Average adult	12+ (adult)	18.74	1.235 (0.945)	3.09 (2.74)	1.03

(avsmalnande delen av var 6-7 cm på de som har en avsmalnande del)

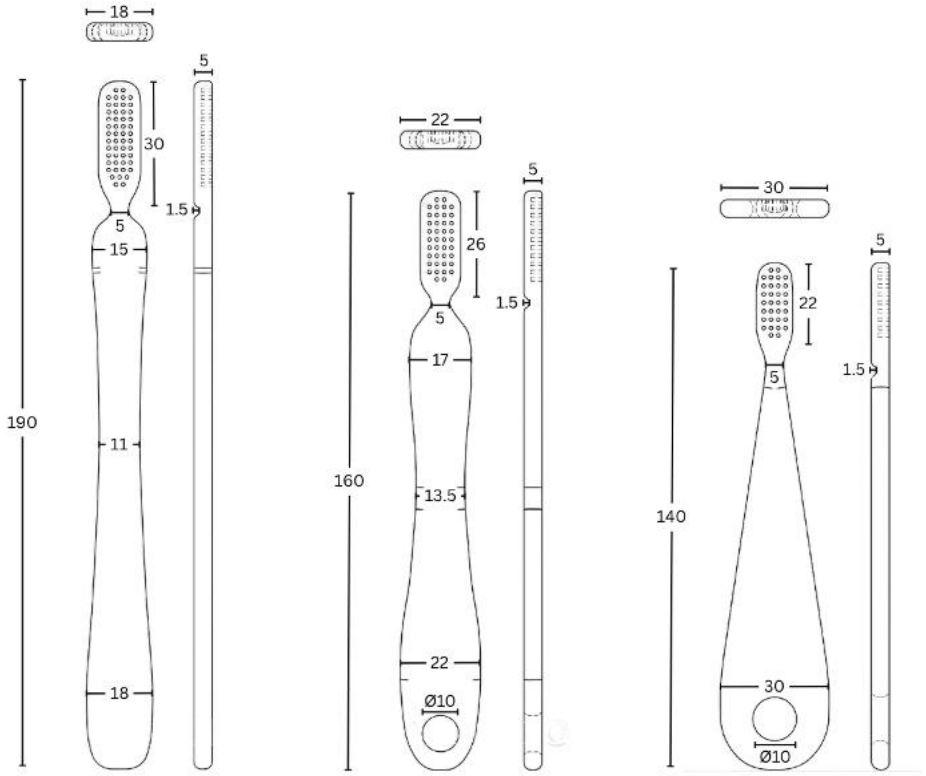
APPENDIX G - Matlab script for groove calculation

```
% GROOVE DEPTH CALCULATION
clear; clc;
%% INPUT PARAMETERS
b = 0.005;          % width (m)
h = 0.005;          % original height (m)
Lt= 0.175;          % distance teeth reaction force-end
Lg= 0.025;          % distance teeth reaction force-groove
Sf = 5;              % safety factor
R_teeth= 2;         % highest brushing force (N)
F_break = R_teeth*Sf;  % intended breaking force (N)
% Source brushing force:  DOI: 10.1007/s00784-012-0788-z
sigma_b = 80e6;    % bending strength (Pa) (across fibres)
Kt = 1.5;          % stress concentration factor for round groove
%% BENDING MOMENTS
M = F_break * Lg; %Bending moment at the groove
%sigma = Kt * M / W
%W = (b * (h*x)^2) / 6
%% SOLVE FOR x
x = sqrt((Kt * 6 * M) / (b * h^2 * sigma_b)) %procentage of original height at
breaking point
groove_depth=h*(1-x)
% GROOVE DEPTH CALCULATION
%_____________________________________________AGE: 6-12
%% INPUT PARAMETERS
b = 0.005;          % width (m)
h = 0.005;          % original height (m)
Lg= 0.023;          % distance teeth reaction force-groove
Sf = 5;              % safety factor
R_teeth= 2;         % highest brushing force (N)
F_break = R_teeth*Sf;  % intended breaking force (N)
% Source brushing force:  DOI: 10.1007/s00784-012-0788-z
sigma_b = 80e6;    % bending strength (Pa) (across fibres)
Kt = 1.5;          % stress concentration factor for round groove
%% BENDING MOMENTS
M = F_break * Lg; %Bending moment at the groove
%sigma = Kt * M / W
%W = (b * (h*x)^2) / 6
%% SOLVE FOR x
x = sqrt((Kt * 6 * M) / (b * h^2 * sigma_b)) %procentage of original height at
breaking point
groove_depth_6to12=h*(1-x)
%_____________________________________________AGE: 6-12
% GROOVE DEPTH CALCULATION
%% INPUT PARAMETERS
b = 0.005;          % width (m)
h = 0.005;          % original height (m)
Lg= 0.021;          % distance teeth reaction force-groove
Sf = 5;              % safety factor
```

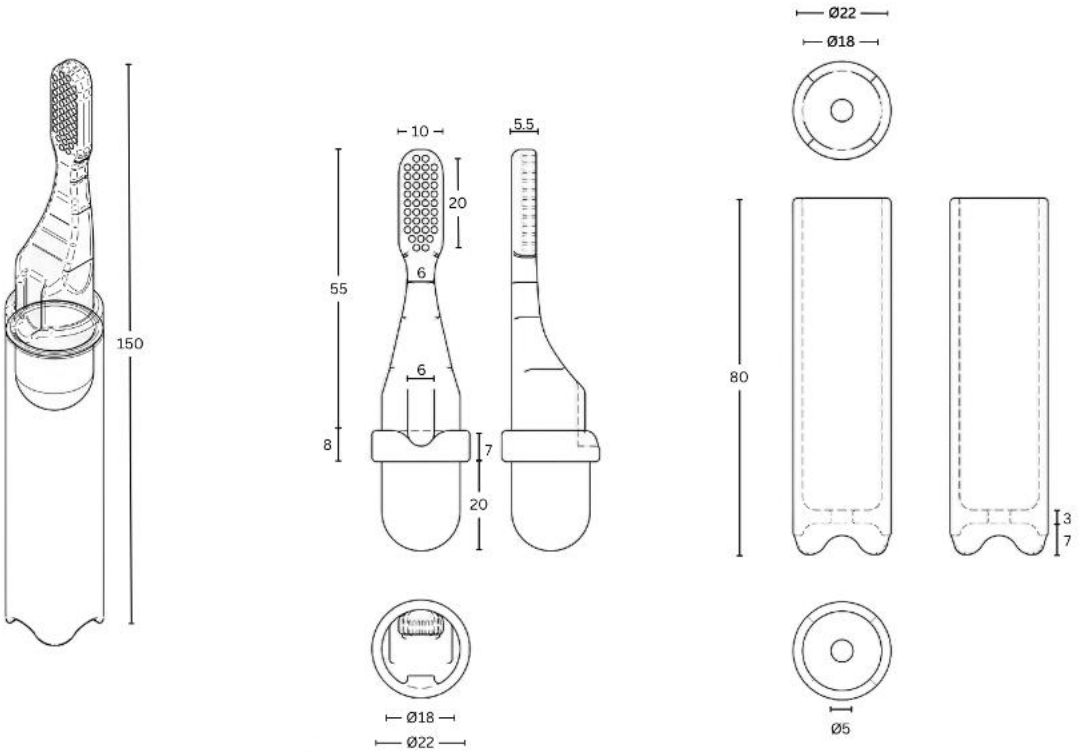
```
R_teeth= 2;         % highest brushing force (N)
F_break = R_teeth*Sf;  % intended breaking force (N)
% Source brushing force:  DOI: 10.1007/s00784-012-0788-z
sigma_b = 80e6;    % bending strength (Pa) (across fibres)
Kt = 1.5;          % stress concentration factor for round groove
%% BENDING MOMENTS
M = F_break * Lg; %Bending moment at the groove
%sigma = Kt * M / W
%W = (b * (h*x)^2) / 6
%% SOLVE FOR x
x = sqrt((Kt * 6 * M) / (b * h^2 * sigma_b)) %procentage of original height at
breaking point
groove_depth_0to6=h*(1-x)
```

APPENDIX H - Measurements for Concept 1 and Concept 2

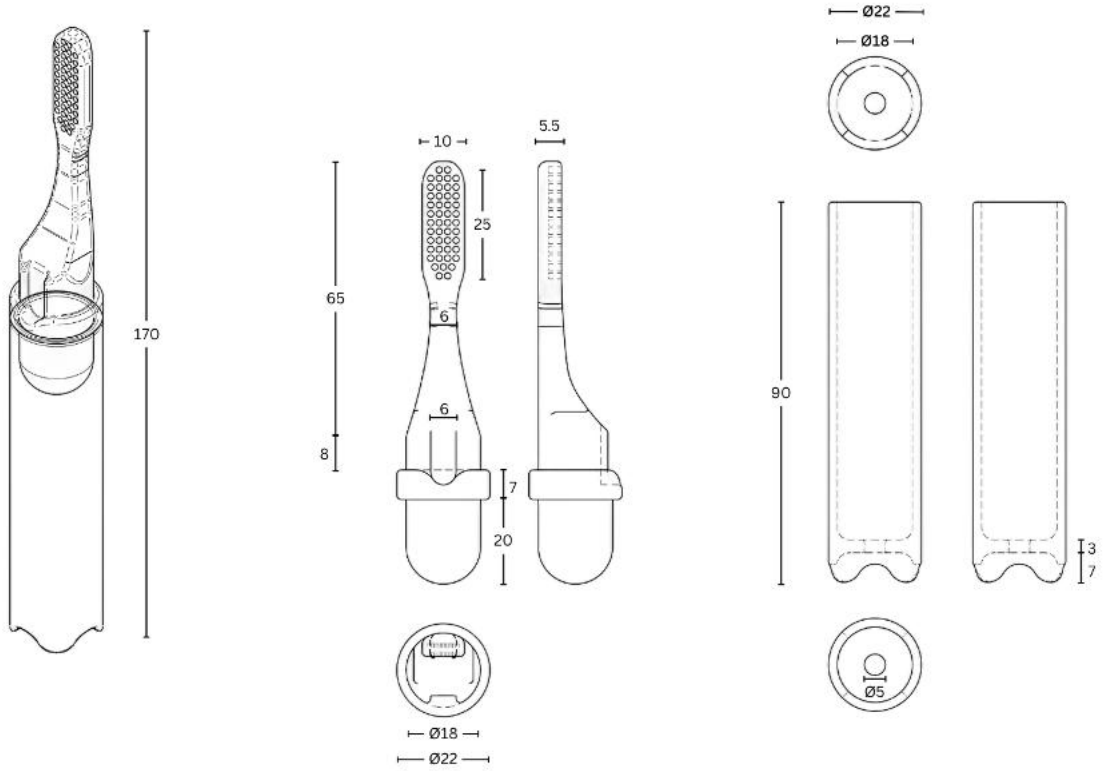
Concept 1



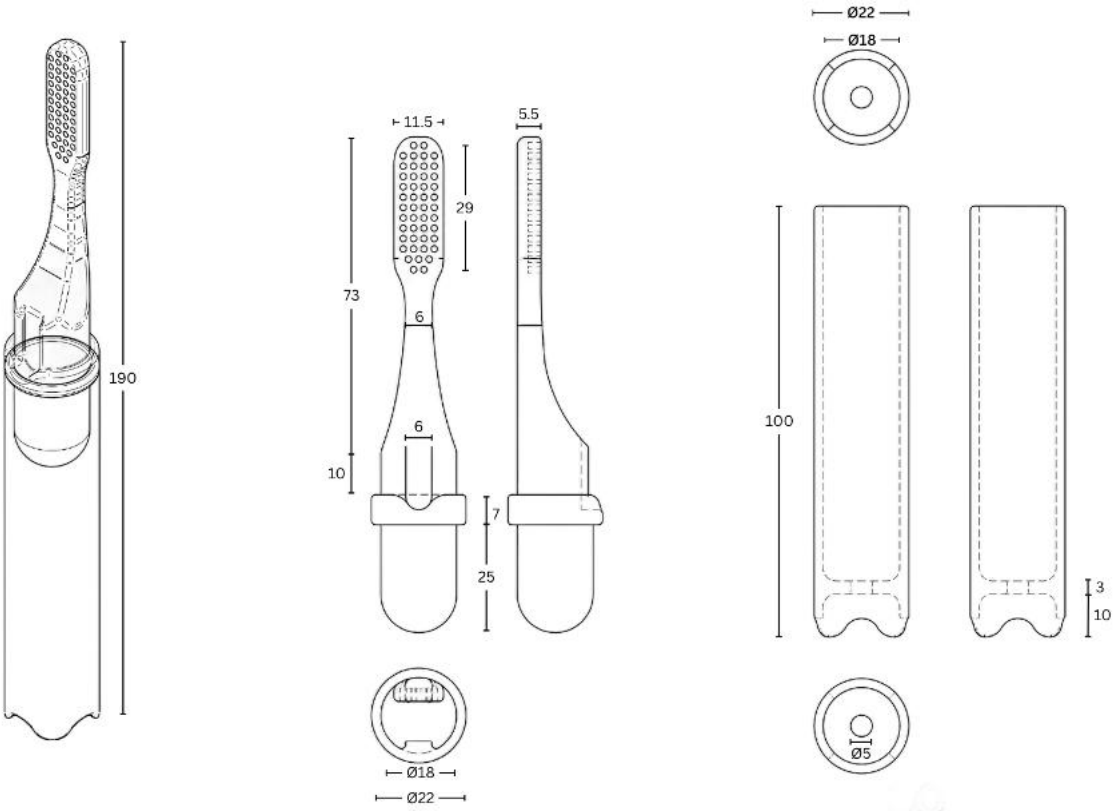
Concept 2, ages 0-6



Concept 2, ages 6-12



Concept 2, ages 12+



APPENDIX I - AI prompts

The following prompts were used for generating context images for the toothbrushes with the help of the Ai generator "Leonardo.Ai":

Concept 1:

- Generate a photo of four toothbrushes hanging in a row, on hooks mounted on a wall in an African public school environment. The wall behind the toothbrushes should have few cracks in the paint, but no holes. there are windows on both sides, barely visible. See image reference 4.

- The toothbrushes are hanging with the hole up and the brush down. The hook is going through the hole.

- The orientation of "izine" and the hand-written text should be the same on all toothbrushes. in reference image 4 it is the wrong way around. text should go from bottom of image to top.

- The order of the color on the bristles should be the same on all toothbrushes as it is in reference 4.

- The photo is shot with a 50mm lens, and only the wall, hooks and toothbrushes are visible in the picture. Natural light. Photo-realistic.

- Each toothbrush has a name written on it.
  The names are:
  Aline
  Jean
  Keza
  Niyo
  These are typed in a childish way with a permanent marker. See the reference picture 1 and 4.

- Reference 1, 2 and 3 are product images. Image reference 1 has two types of toothbrush. Only use the one with a name written on it. Reference 2 is the toothbrush at an angle and 3 is from the front with brush facing downward, as it should be in the rendered image.

Concept 2:

- A 5-year-old child's hand holding the toothbrush in reference image under running water from a large, clean water tank tap. Toothbrush in reference picture is 16 cm. African setting. Red dirt ground. Warm natural light. Soft blurry background with shallow depth of field. Realistic documentary photography style. Focus on the child's hand and toothbrush. Landscape format, 16:9 ratio.

Packaging mockup:

Create a realistic premium product mockup of four kraft paper pouch packages for toothbrushes. The photo is a flatlay, taken from above with the packages laying down. IMPORTANT: this is a soft kraft paper bag/pouch package, NOT a cardboard box.

Use the uploaded reference image 2 for the exact pouch shape, proportions, folds, sealing style, and overall packaging structure.
Use the uploaded reference image 1 as the exact printed front design for the four different pouches.

Realistic brown kraft paper texture with visible paper fibers, soft creases, matte finish, and authentic packaging details.

Professional commercial product photography aesthetic. Soft studio lighting, subtle shadows, clean composition, photorealistic materials, ultra detailed texture, realistic print application on kraft paper.

Camera angle: front-facing with a slight 3/4 perspective. Clean neutral beige or off-white studio background. Focus on elegant sustainable packaging design.
.

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